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Generate floor response spectra: Part 1. Direct spectra-to-spectra method

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HIGHLIGHTS

- The concept of tRS is proposed to deal with tuning of equipment and structures.
- Established statistical approaches for estimating tRS corresponding to given GRS.
- Derived a new modal combination rule from the theory of random vibration.
- Developed efficient & accurate direct method for generating floor response spectra.

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ABSTRACT

A direct spectra-to-spectra method is developed for generating floor response spectra (FRS), which are used as input for seismic design and assessment of secondary systems in nuclear facilities. Only ground response spectra (GRS) and basic modal information of primary structures, including natural frequencies, modal damping, participation factors, and mode shapes, which can be readily obtained from a modal analysis, are needed. In tuning cases, when equipment is resonant with the primary structure, the concept of t-response spectrum (tRS) is proposed, and statistical relationships between tRS and GRS developed in the companion paper are employed to determine FRS. Furthermore, a new modal combination rule (called FRS-CQC), which considers the correlation between modal responses of a structure and correlation between responses of equipment and its supporting structure, is derived based on random vibration theory. Numerical examples of a typical service building in nuclear power plants indicate that results from the proposed method agree extremely well with the numerically "exact" FRS obtained from a large number of sets of time history analyses. It is observed that FRS from time history analyses have large variabilities, particularly in tuning cases. Numerical examples demonstrate that the modal combination rules have a significantly effect on the accuracy of FRS by the direct spectra-to-spectra method. The proposed direct spectra-to-spectra method, which avoids the deficiencies of time history methods, is of excellent accuracy, efficiency, and simplicity; it is convenient to implement in practice for generating FRS.

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1. Introduction

1.1. Floor response spectra

ASCE Standard 4-98 (1998) recommends that floor response spectra (FRS) can be generated by time history analyses or a direct spectra-to-spectra method, as illustrated in Fig. 1. However, it has been recognized that there are significant variabilities in FRS generated by the time history method, in the sense that two spectrum-compatible time histories may give significantly different FRS, as shown in Section 5. Hence, if only a single set of tri-directional earthquake time histories, or even a small number of sets of tri-directional earthquake time histories, is used in the time history analysis to calculate FRS, the FRS obtained is not reliable. Consequently, a large number of sets of

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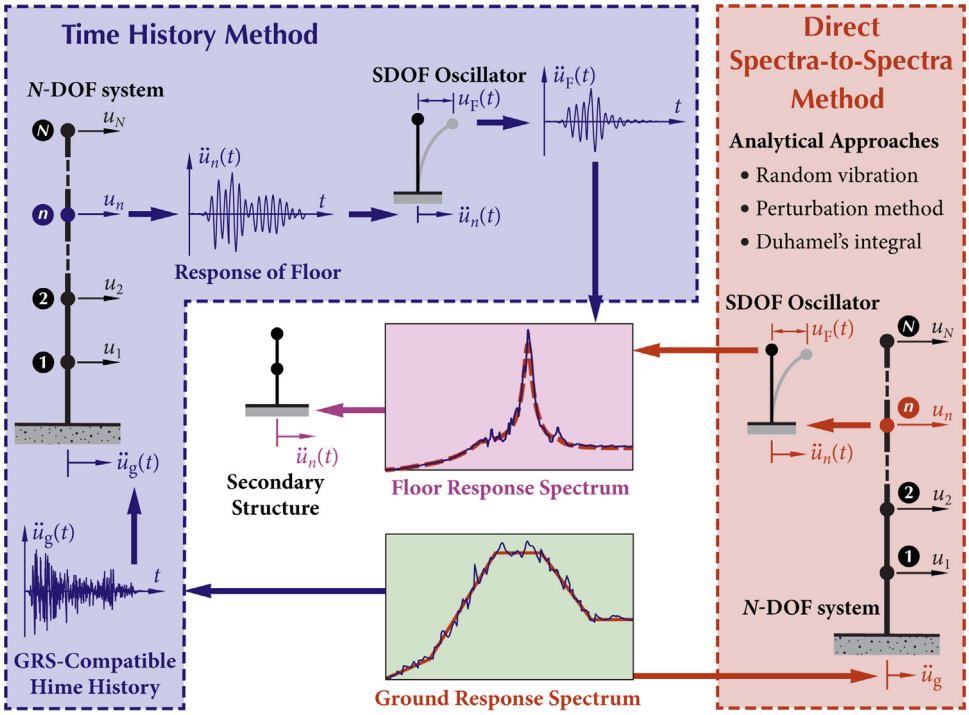


Fig. 1. Two methods of generating floor response spectra.

tri-directional earthquake time histories may be required to obtain an reliable probabilistic description of FRS; but this procedure is not only cumbersome but also computationally expensive.

Various direct spectra-to-spectra methods have been developed to avoid the deficiencies of time history methods. In direct methods, FRS are analytically expressed in terms of ground motion spectra and some basic modal information of the primary structure, including modal frequencies, modal shapes, and modal participation factors, which can be obtained from a modal analysis. Some methods are developed under a decouple analysis, which assumes that the interaction effect between equipment and the supporting structure is negligible. Duff (1975) developed a simplified graphical method to generate FRS directly from GRS. However, the amplification factors defined in this method depend on the so-called pseudo ground damping which is determined empirically. Singh (1975, 1980) proposed a direct method based on random vibration theory. However, this method is not applicable when equipment is resonant with multiple structural modes for structures with closely-spaced vibration modes. Yasui et al. (1993) and An et al. (2013) derived approximate analytical results of FRS in time domain instead of frequency domain. An et al. (2013) were the first to evaluate analytically the convolution between two unit impulse response functions. However, they used conventional combination methods, such as the SRSS (Square Root of Sum of Squares) and CQC (Complete Quadratic Combination) rules (Der Kiureghian, 1980, 1981), for modal combination; there may be large errors in the resulting FRS, particularly in tuning cases and when the supporting structure has closely-spaced modes.

Other methods were developed considering a combined primary–secondary systems, in which the effect of equipment-structure interaction is considered. Sackman et al. (1983), Der Kiureghian et al. (1983) and Igusa and Der Kiureghian (1985) employed a perturbation technique for modal analysis of the combined system to generate FRS.

In current practice, FRS is generally generated without considering the effect interaction. For the tuning cases, one of the terms becomes the first-order derivative of the acceleration response spectrum with respect to damping (see Eq. (46)), which cannot be determined analytically in terms of the input GRS. Various methods (An et al., 2013; Der Kiureghian et al., 1983; Igusa and Der Kiureghian, 1985; Singh, 1980; Yasui et al., 1993) have been applied to obtain an approximate value of this tuning term.

Although there have been a number of direct spectra-to-spectra methods available for generating FRS since 1970s, they have not been widely used in nuclear industry, because these methods give conservative results in some frequency ranges but unconservative results in other frequency ranges due to the various approximations used. Furthermore, it is unknown when and by how much the FRS obtained is conservative or unconservative. If the results are scaled so that the resulting FRS are conservative over the entire frequency range, FRS in some frequency ranges become excessively conservative. The results of FRS should be conservative but not too conservative; otherwise, the cost of seismic qualification of systems, structures, and components (SSC) of nuclear power plants will be significantly increased.

The objective of this paper is to develop an accurate direct spectra-to-spectra method. To achieve this objective, there are two major novel contributions in this study:

1 The concept of t-response spectrum (tRS) is proposed to deal with tuning cases. Statistical relationships between tRS and GRS are developed in the companion paper based on simulation results from a large number of real earthquake records.
2 A new combination rule, called FRS-CQC (Complete Quadratic Combination), is derived from the theory of random vibration. FRS-CQC accounts for not only correlation between responses of structural modes but also correlations between responses of equipment and structural modes. FRS-CQC is capable of dealing with structures with closely spaced modes.

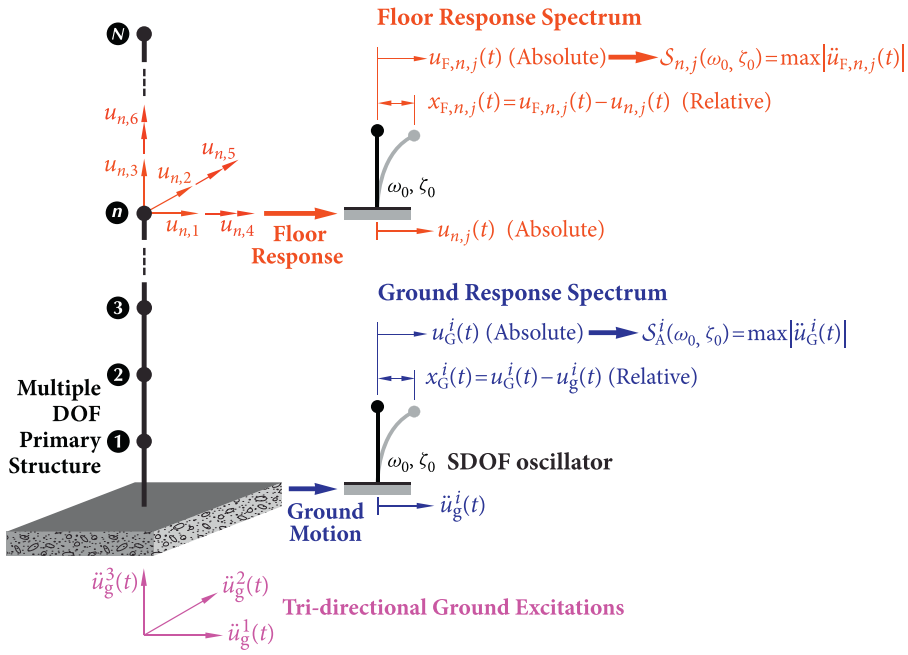


Fig. 2. Response spectra.

In Section 2, the general formulation of FRS of a three-dimensional structure under the excitation of tri-directional ground motions is presented. The direct spectra-to-spectra method for generating FRS is developed in Section 3, considering tuning and non-tuning cases, and modal combination for multiple degrees-of-freedom (DOF) structures. Numerical examples of FRS for a service building of a nuclear power plant are presented in Section 4 to validate and demonstrate the high accuracy and efficiency of the proposed method. Some conclusions are drawn in Section 5.

2. Response spectra

2.1. Ground response spectra

In seismic design and assessment of nuclear power plants, seismic excitations in two orthogonal horizontal directions H_1 and H_2, and vertical direction V are usually applied. Suppose that $u_g^i(t)$, $i = 1, 2, 3$, where $u_g^1(t) = u_g^{H_1}(t)$, $u_g^2(t) = u_g^{H_2}(t)$, and $u_g^3(t) = u_g^V(t)$, is the displacement of the ground motion in direction i. When a single degree-of-freedom (SDOF) oscillator with circular frequency ω_0 and damping coefficient ζ_0 is subjected to this ground motion, as shown in Fig. 2, the equation of motion is

$$\ddot{x}_G^i + 2\zeta_0\omega_0\dot{x}_G^i + \omega_0^2 x_G^i = -\ddot{u}_g^i(t), \tag{1}$$

where $x_G^i(t) = u_G^i(t) - u_g^i(t)$ is the relative displacement of the oscillator and $u_G^i(t)$ is the absolute displacement. The subscript "G" denotes that the oscillator is mounted on the ground. The absolute acceleration is $\ddot{u}_G^i(t) = \ddot{x}_G^i(t) + \ddot{u}_g^i(t) = -(2\zeta_0\omega_0\dot{x}_G^i + \omega_0^2 x_G^i)$. The maximum absolute acceleration of the oscillator $\mathcal{S}_A^i(\omega_0, \zeta_0) = \max|\ddot{u}_G^i(t)|$ is the ground (acceleration) response spectrum (GRS) in direction i.

2.2. FRS of SDOF primary structure

For the special case when the primary structure is SDOF with circular frequency ω and damping coefficient ζ, $u(t)$ and $x(t) = u(t) - u_g(t)$ are the absolute and relative displacements of the structure, respectively, satisfying

$$\ddot{x}(t) + 2\zeta\omega\dot{x}(t) + \omega^2 x(t) = -\ddot{u}_g(t), \tag{2}$$

$$\ddot{u}(t) = \ddot{x}(t) + \ddot{u}_g(t) = -2\zeta\omega\dot{x}(t) - \omega^2 x(t). \tag{3}$$

The motion of a SDOF oscillator with circular natural frequency ω_0 and damping coefficient ζ_0 mounted on the primary structure (Fig. 3) is governed by

$$\ddot{x}_F + 2\zeta_0\omega_0\dot{x}_F + \omega_0^2 x_F = -\ddot{u}(t), \tag{4}$$

$$\ddot{u}_F(t) = \ddot{x}_F(t) + \ddot{u}(t) = -2\zeta_0\omega_0\dot{x}_F(t) - \omega_0^2 x_F(t), \tag{5}$$

where $x_F(t) = u_F(t) - u(t)$ and $u_F(t)$ are the relative and absolute displacements of the oscillator. The maximum absolute acceleration of the oscillator $\mathcal{S}_F(\omega_0, \zeta_0) = \max|\ddot{u}_F(t)|$ is the floor (acceleration) response spectrum (FRS) of the SDOF primary structure.

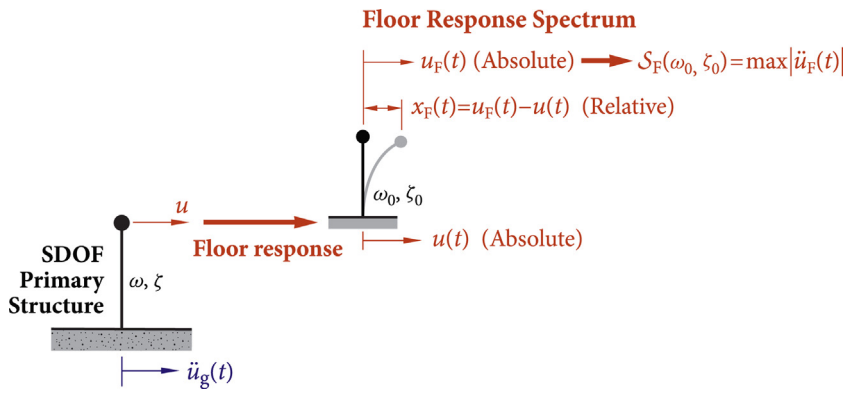


Fig. 3. FRS of SDOF primary structure.

2.3. FRS of multiple DOF primary structure

Consider a three-dimensional model of a structure with N nodes. A typical node N has six DOF: three translational DOF $u_{n,1}$, $u_{n,2}$, $u_{n,3}$, and three rotational DOF $u_{n,4}$, $u_{n,5}$, $u_{n,6}$. The structure is subjected to tri-directional seismic excitations (Fig. 2). The relative displacement vector $\mathbf{x}$ of dimension $6N$ is governed by

$$\mathbf{M}\ddot{\mathbf{x}}(t)+\mathbf{C}\dot{\mathbf{x}}(t)+\mathbf{K}\mathbf{x}(t)=-\mathbf{M}\sum_{i=1}^{3}\mathcal{I}^i\ddot{u}_g^i(t), \tag{6}$$

where

$$\mathbf{x}=\begin{Bmatrix}\mathbf{x}_1\\ \mathbf{x}_2\\ \vdots\\ \mathbf{x}_N\end{Bmatrix},\quad \mathbf{x}_n=\begin{Bmatrix}x_{n,1}\\ x_{n,2}\\ \vdots\\ x_{n,6}\end{Bmatrix},\quad \mathcal{I}^i=\begin{Bmatrix}\mathbf{1}^i\\ \mathbf{1}^i\\ \vdots\\ \mathbf{1}^i\end{Bmatrix},\quad \mathbf{1}^i=\begin{Bmatrix}\delta_{i1}\\ \delta_{i2}\\ \vdots\\ \delta_{i6}\end{Bmatrix},$$

$\mathbf{M}$, $\mathbf{C}$, $\mathbf{K}$ are, respectively, the mass, damping, and stiffness matrices of dimension $6N\times 6N$, $\mathbf{x}_n$ is the relative displacement vector of node N, $\mathcal{I}^i$ is the influence vector of the seismic excitation in direction i, and δ_{ij} denotes the Kronecker delta function.

Let $\mathbf{x}=\mathbf{x}^1+\mathbf{x}^2+\mathbf{x}^3$, where $\mathbf{x}^i$ is the relative displacement vector due to the earthquake excitation $u_g^i(t)$ in direction i. Hence, $x_{n,j}^i=u_{n,j}^i-u_g^i\delta_{ij}$, where $x_{n,j}^i$ and $u_{n,j}^i$ are, respectively, the relative and absolute displacements of node N in direction j due to the earthquake excitation in direction i. Since the system is linear, from Eq. (6), $\mathbf{x}^i$ is governed by

$$\mathbf{M}\ddot{\mathbf{x}}^i(t)+\mathbf{C}\dot{\mathbf{x}}^i(t)+\mathbf{K}\mathbf{x}^i(t)=-\mathbf{M}\mathcal{I}^i\ddot{u}_g^i(t),\quad i=1,2,3. \tag{7}$$

Free vibration. Consider the undamped free vibration $\mathbf{M}\ddot{\mathbf{x}}(t)+\mathbf{K}\mathbf{x}(t)=\mathbf{0}$. Let $\omega_1,\omega_2,\ldots,\omega_{6N}$ be the $6N$ natural frequencies and $\boldsymbol{\Phi}=[\boldsymbol{\varphi}_1,\boldsymbol{\varphi}_2,\ldots,\boldsymbol{\varphi}_{6N}]$ be the modal matrix, where $\boldsymbol{\varphi}_k=\{\boldsymbol{\varphi}_{1,k}^T,\boldsymbol{\varphi}_{2,k}^T,\ldots,\boldsymbol{\varphi}_{N,k}^T\}^T$ is the mode shape of the kth mode, with $\boldsymbol{\varphi}_{n,k}=\{\varphi_{n,1;k},\varphi_{n,2;k},\ldots,\varphi_{n,6;k}\}^T$. In element $\varphi_{n,j;k}$, the first subscript N refers to the node number, the second subscript j indicates the direction of response, and the third subscript k refers to the mode number.

The modal matrix $\boldsymbol{\Phi}$ has the following orthogonal properties

$$\begin{aligned}\boldsymbol{\Phi}^T\mathbf{M}\boldsymbol{\Phi}&=\mathrm{diag}\{\bar{m}_1,\bar{m}_2,\ldots,\bar{m}_{6N}\}^T=\bar{\mathbf{m}},\\ \boldsymbol{\Phi}^T\mathbf{K}\boldsymbol{\Phi}&=\bar{\mathbf{m}}\boldsymbol{\Omega}^2=\mathrm{diag}\{\bar{m}_1\omega_1^2,\bar{m}_2\omega_2^2,\ldots,\bar{m}_{6N}\omega_{6N}^2\}^T,\\ \boldsymbol{\Omega}&=\mathrm{diag}\{\omega_1,\omega_2,\ldots,\omega_{6N}\}^T,\end{aligned} \tag{8}$$

where $\bar{m}_1,\bar{m}_2,\ldots,\bar{m}_{6N}$ are the modal masses. Assume that the structure has classical damping so that the modal matrix $\boldsymbol{\Phi}$ can also diagonalize the damping matrix, i.e.,

$$\boldsymbol{\Phi}^T\mathbf{C}\boldsymbol{\Phi}=\mathrm{diag}\{\bar{c}_1,\bar{c}_2,\ldots,\bar{c}_{6N}\}^T,\qquad \bar{c}_k=\bar{m}_k\cdot 2\zeta_k\omega_k. \tag{9}$$

Forced vibration. Apply the transformation

$$\mathbf{x}^i(t)=\boldsymbol{\Phi}\mathbf{Q}^i(t),\quad i=1,2,3, \tag{10}$$

where

$$\mathbf{Q}^i = \begin{Bmatrix} \mathbf{Q}_1^i \\ \mathbf{Q}_2^i \\ \vdots \\ \mathbf{Q}_N^i \end{Bmatrix} = \begin{Bmatrix} \Gamma_1^i q_1^i \\ \Gamma_2^i q_2^i \\ \vdots \\ \Gamma_{6N}^i q_{6N}^i \end{Bmatrix}, \quad \mathbf{Q}_n^i = \begin{Bmatrix} \Gamma_{n,1}^i q_{n,1}^i \\ \Gamma_{n,2}^i q_{n,2}^i \\ \vdots \\ \Gamma_{n,6}^i q_{n,6}^i \end{Bmatrix}, \quad \mathbf{x}^i = \begin{Bmatrix} \mathbf{x}_1^i \\ \mathbf{x}_2^i \\ \vdots \\ \mathbf{x}_N^i \end{Bmatrix}, \quad \mathbf{x}_n^i = \begin{Bmatrix} x_{n,1}^i \\ x_{n,2}^i \\ \vdots \\ x_{n,6}^i \end{Bmatrix}, \tag{11}$$

$$x_{n,j}^i = \sum_{\nu=1}^{N} \sum_{\delta=1}^{6} \varphi_{n,j;6(\nu-1)+\delta} \Gamma_{\nu,\delta}^i q_{\nu,\delta}^i = \sum_{k=1}^{6N} \varphi_{n,j;k} \Gamma_k^i q_k^i, \tag{12}$$

$$\mathrm{L}_{n,j}^i = \boldsymbol{\varphi}_{6(n-1)+j}^T \mathbf{M} \mathcal{I}^i, \quad \text{or} \quad \mathrm{L}_k^i = \boldsymbol{\varphi}_k^T \mathbf{M} \mathcal{I}^i, \tag{13}$$

$$\Gamma_{n,j}^i = \frac{\mathrm{L}_{n,j}^i}{\bar{m}_{6(n-1)+j}}, \quad \text{or} \quad \Gamma_k^i = \frac{\mathrm{L}_k^i}{\bar{m}_k} = \frac{\boldsymbol{\varphi}_k^T \mathbf{M} \mathcal{I}^i}{\boldsymbol{\varphi}_k^T \mathbf{M} \boldsymbol{\varphi}_k}. \tag{14}$$

For ease of presentation, the two-subscript-notation n, j (node, direction) and the one-subscript-notation $k = 6(n-1)+j$ are used interchangeably; the former is advantageous in describing the meaning of the quantity in terms of node and direction, and the latter gives the position of the quantity in the corresponding vector. $\mathrm{L}_{n,j}^i$ is the *earthquake excitation factor*, quantifying the contribution of earthquake excitation in the ith direction to the modal response $q_{n,j}^i$. Γ_k^i is the *modal participation factors*; if Γ_k^i is small, then the contribution of mode $\boldsymbol{\varphi}_k$ to the structural response due to excitation in the ith direction is small.

Substituting Eq. (10) into (7) and multiplying $\boldsymbol{\Phi}^T$ from the left yield

$$(\boldsymbol{\Phi}^T \mathbf{M} \boldsymbol{\Phi}) \ddot{\mathbf{Q}}^i(t) + (\boldsymbol{\Phi}^T \mathbf{C} \boldsymbol{\Phi}) \dot{\mathbf{Q}}^i(t) + (\boldsymbol{\Phi}^T \mathbf{K} \boldsymbol{\Phi}) \mathbf{Q}^i(t) = -\boldsymbol{\Phi}^T \mathbf{M} \mathcal{I}^i \ddot{u}_g^i(t).$$

Using relations (8), (9), (13) and (14) gives

$$\ddot{q}_k^i(t) + 2\zeta_k \omega_k \dot{q}_k^i(t) + \omega_k^2 q_k^i(t) = -\ddot{u}_g^i(t), \qquad k = 1, 2, \ldots, 6N, \qquad i = 1, 2, 3. \tag{15}$$

The absolute acceleration of the Nth node in direction j due to the earthquake excitation in direction i can be obtained using Eqs. (12) and (15)

$$\ddot{u}_{n,j}^i(t) = \ddot{x}_{n,j}^i(t) + \ddot{u}_g^i(t)\delta_{ij} = \sum_{k=1}^{6N} \varphi_{n,j;k} \Gamma_k^i \ddot{q}_k^i(t) + \ddot{u}_g^i(t)\delta_{ij} = \sum_{k=1}^{6N} \varphi_{n,j;k} \Gamma_k^i [-\ddot{u}_g^i(t) - (2\zeta_k \omega_k \dot{q}_k^i + \omega_k^2 q_k^i)] + \ddot{u}_g^i(t)\delta_{ij},$$

$$\sum_{k=1}^{6N} \varphi_{n,j;k} \Gamma_k^i = \delta_{ij} = -\sum_{k=1}^{6N} \varphi_{n,j;k} \Gamma_k^i (2\zeta_k \omega_k \dot{q}_k^i + \omega_k^2 q_k^i) = -\sum_{k=1}^{6N} \ddot{u}_{n,j;k}^i, \qquad \ddot{u}_{n,j;k}^i = \varphi_{n,j;k} \Gamma_k^i (2\zeta_k \omega_k \dot{q}_k^i + \omega_k^2 q_k^i), \tag{16}$$

in which $\ddot{u}_{n,j;k}^i$ is the contribution from the kth mode and $\varphi_{n,j;k}\Gamma_k^i$ is the contribution factor.

Floor response spectrum. If the absolute response $u_{n,j}^i(t)$ of the Nth node in direction j due to earthquake excitation in direction i is input to a SDOF oscillator with circular frequency ω_0 and damping coefficient ζ_0, as shown in Fig. 2, the governing equation of motion is

$$\ddot{x}_{\mathrm{F},n,j}^i + 2\zeta_0 \omega_0 \dot{x}_{\mathrm{F},n,j}^i + \omega_0^2 x_{\mathrm{F},n,j}^i = -\ddot{u}_{n,j}^i(t), \tag{17}$$

$$\ddot{u}_{\mathrm{F},n,j}^i(t) = \ddot{x}_{\mathrm{F},n,j}^i(t) + \ddot{u}_{n,j}^i(t) = -2\zeta_0 \omega_0 \dot{x}_{\mathrm{F},n,j}^i - \omega_0^2 x_{\mathrm{F},n,j}^i, \tag{18}$$

where $x_{\mathrm{F},n,j}^i(t) = u_{\mathrm{F},n,j}^i(t) - u_{n,j}^i(t)$ is the displacement of the oscillator relative to the Nth node in direction j, and $u_{\mathrm{F},n,j}^i(t)$ is the absolute displacement of the oscillator. The subscript "F" denotes that the oscillator is mounted on the floor. The maximum absolute acceleration of the oscillator

$$\mathcal{S}_{n,j}^i(\omega_0, \zeta_0) = |\ddot{u}_{\mathrm{F},n,j}^i(t)|_{\max} \tag{19}$$

is the floor (acceleration) response spectrum (FRS) of the Nth node (floor) in direction j subjected to earthquake excitation in direction i.

It is specified in ASCE 4-98 (1998) that, for direct spectra-to-spectra methods, when the response spectrum of a given direction at a given location has contributions from more than one spatial component of earthquakes, these contributions shall be combined by the SRSS rule. Hence, combining contributions from tri-directional earthquake excitations, FRS of the Nth node in direction j is given by

$$\mathcal{S}_{n,j}(\omega_0, \zeta_0) = \sqrt{\sum_{i=1}^{3} [\mathcal{S}_{n,j}^i(\omega_0, \zeta_0)]^2}. \tag{20}$$

3. Direct method for generating FRS

In this section, a direct spectra-to-spectra method for generating FRS is developed based on Duhamel's integral.

3.1. SDOF oscillator mounted on SDOF structure

Consider a SDOF oscillator mounted on a SDOF structure, as shown in Fig. 3. Adopt the notations

$$h(t) = \mathrm{e}^{-\zeta\omega t}\,\frac{\sin\omega_{\mathrm{d}}t}{\omega_{\mathrm{d}}},\quad h^{\mathrm{c}}(t) = \mathrm{e}^{-\zeta\omega t}\,\frac{\cos\omega_{\mathrm{d}}t}{\omega_{\mathrm{d}}},\quad \omega_{\mathrm{d}} = \omega\sqrt{1-\zeta^2}, \tag{21}$$

$$h_0(t) = \mathrm{e}^{-\zeta_0\omega_0 t}\,\frac{\sin\omega_{0,\mathrm{d}}t}{\omega_{0,\mathrm{d}}},\quad h_0^{\mathrm{c}}(t) = \mathrm{e}^{-\zeta_0\omega_0 t}\,\frac{\cos\omega_{0,\mathrm{d}}t}{\omega_{0,\mathrm{d}}},\quad \omega_{0,\mathrm{d}} = \omega_0\sqrt{1-\zeta_0^2}. \tag{22}$$

Motion of structure. For SDOF system (2) with zero initial conditions, using Duhamel's integral (Clough and Penzien, 1975), the relative displacement $x(t)$ and the relative velocity $\dot{x}(t)$ can be expressed as

$$x(t) = h(t) * \ddot{u}_{\mathrm{g}}(t),\qquad \dot{x}(t) = \dot{h}(t) * \ddot{u}_{\mathrm{g}}(t), \tag{23}$$

where $h(t)$ is the unit impulse response function of the structure and ω_{d} is the damped circular frequency defined by Eq. (21). The derivative of $h(t)$ is

$$\dot{h}(t) = -\frac{\zeta}{\sqrt{1-\zeta^2}}\mathrm{e}^{-\zeta\omega t}\sin\omega_{\mathrm{d}}t + \mathrm{e}^{-\zeta\omega t}\cos\omega_{\mathrm{d}}t = -\zeta\omega h(t) + \mathrm{e}^{-\zeta\omega t}\cos\omega_{\mathrm{d}}t. \tag{24}$$

Substituting Eq. (23) into (3), the absolute floor acceleration of the structure is given by

$$\ddot{u}(t) = -2\zeta\omega\dot{h}(t) * \ddot{u}_{\mathrm{g}}(t) - \omega^2 h(t) * \ddot{u}_{\mathrm{g}}(t). \tag{25}$$

Motion of oscillator. The motion of the structure, to which the oscillator is attached, defines the input to the SDOF oscillator with circular natural frequency ω_0 and damping coefficient ζ_0; the relative and absolute motions of the oscillator are governed by Eqs. (4) and (5), respectively. Using Duhamel's integral and Eq. (25), the relative displacement $x_{\mathrm{F}}(t)$ and velocity $\dot{x}_{\mathrm{F}}(t)$ between the structure and the oscillator are

$$\begin{aligned} x_{\mathrm{F}}(t) &= h_0(t) * \ddot{u}(t) = -2\zeta\omega h_0(t) * \dot{h}(t) * \ddot{u}_{\mathrm{g}}(t) - \omega^2 h_0(t) * h(t) * \ddot{u}_{\mathrm{g}}(t),\\ \dot{x}_{\mathrm{F}}(t) &= \dot{h}_0(t) * \ddot{u}(t) = -2\zeta\omega\dot{h}_0(t) * \dot{h}(t) * \ddot{u}_{\mathrm{g}}(t) - \omega^2\dot{h}_0(t) * h(t) * \ddot{u}_{\mathrm{g}}(t), \end{aligned} \tag{26}$$

where the unit impulse response function $h_0(t)$ is defined by Eq. (22).

Substituting Eq. (26) into (5), the absolute acceleration of the oscillator is expressed as

$$\begin{aligned} \ddot{u}_{\mathrm{F}}(t) &= -2\zeta_0\omega_0\dot{x}_{\mathrm{F}}(t) - \omega_0^2 x_{\mathrm{F}}(t) = [4\zeta_0\zeta\omega_0\omega\cdot\dot{h}_0(t) * \dot{h}(t) + 2\zeta_0\omega_0\omega^2\cdot\dot{h}_0(t) * h(t) + 2\zeta\omega_0^2\omega\cdot h_0(t) * \dot{h}(t) + \omega_0^2\omega^2\cdot h_0(t) * h(t)] * \ddot{u}_{\mathrm{g}}(t)\\ &= [\,(1-2\zeta_0^2-2\zeta^2+4\zeta_0^2\zeta^2)\omega_0^2\omega^2\cdot h_0(t) * h(t) + 4\zeta_0\zeta\sqrt{(1-\zeta^2)(1-\zeta_0^2)}\omega_0^2\omega^2\cdot h_0^{\mathrm{c}}(t) * h^{\mathrm{c}}(t)\\ &\quad + 2\zeta_0\sqrt{1-\zeta_0^2}(1-2\zeta^2)\omega_0^2\omega^2\cdot h(t) * h_0^{\mathrm{c}}(t) + 2(1-2\zeta_0^2)\zeta\sqrt{1-\zeta^2}\omega_0^2\omega^2\cdot h_0(t) * h^{\mathrm{c}}(t)\,] * \ddot{u}_{\mathrm{g}}(t), \end{aligned} \tag{27}$$

where $h^{\mathrm{c}}(t)$ and $h_0^{\mathrm{c}}(t)$ are defined in Eqs. (21) and (22).

For most SSCs in nuclear power plants, the damping coefficients ζ, $\zeta_0 < 0.1$. When t is sufficiently long, it is reasonable to assume that

$$|h_0(t) * h(t) * \ddot{u}_{\mathrm{g}}(t)|_{\max} \approx |h_0^{\mathrm{c}}(t) * h^{\mathrm{c}}(t) * \ddot{u}_{\mathrm{g}}(t)|_{\max} \approx |h(t) * h_0^{\mathrm{c}}(t) * \ddot{u}_{\mathrm{g}}(t)|_{\max} \approx |h_0(t) * h^{\mathrm{c}}(t) * \ddot{u}_{\mathrm{g}}(t)|_{\max}. \tag{28}$$

In general, the maximum values of the terms in (27) do not occur simultaneously because of the phase differences between the sine and cosine terms. For lightly-damped systems, the values of ζ^2, ζ_0^2, and $\zeta_0\zeta$ are very small compared to 1, so that the corresponding terms are negligible. The maximum response of the oscillator is then reduced to

$$|\ddot{u}_{\mathrm{F}}(t)|_{\max} \approx \omega_0^2\omega^2|h_0(t) * h(t) * \ddot{u}_{\mathrm{g}}(t)|_{\max}, \tag{29}$$

which is expressed analytically as a double convolution. Note that, if the SDOF oscillator is mounted directly on the ground, the term $\omega^2 h(t)$ is removed from Eq. (29) and FRS reduces to GRS, i.e.,

$$\mathcal{S}_{\mathrm{A}}(\omega_0,\zeta_0) = \omega_0^2|h_0(t) * \ddot{u}_{\mathrm{g}}(t)|_{\max} = \left|\omega_0\mathrm{e}^{-\zeta_0\omega t}\sin\omega_0 t * \ddot{u}_{\mathrm{g}}(t)\right|_{\max}. \tag{30}$$

Denote $\mathrm{C}(t) = \mathrm{h}_0(t) * \mathrm{h}(t)$. From the definition of Duhamel's integral, it is obvious that $\mathrm{C}(t)$ is the response of an oscillator with the circular frequency ω_0 and damping coefficient ζ_0 under the excitation of $h(t)$. The equation of motion is written as

$$\ddot{\mathrm{C}}(t) + 2\zeta_0\omega_0\dot{\mathrm{C}}(t) + \omega_0^2\mathrm{C}(t) = h(t) = \frac{1}{\omega_{\mathrm{d}}}\mathrm{e}^{-\zeta\omega}\sin\omega_{\mathrm{d}}t. \tag{31}$$

The general solution for this differential equation is $\mathrm{C}(t) = \mathrm{C_C}(t) + \mathrm{C_P}(t)$, where

$$\mathrm{C_C}(t) = \mathrm{e}^{-\zeta_0\omega_0 t}(C_1\cos\omega_{0,\mathrm{d}}t + C_2\sin\omega_{0,\mathrm{d}}t),\quad \text{for}\quad \zeta_0 < 1, \tag{32}$$

is the complementary solution with coefficients C_1 and C_2 determined by the initial conditions, and $\mathrm{C_P}(t)$ is a particular solution determined in the following.

3.2. Non-tuning case

If $\omega \neq \omega_0$ and $\zeta \neq \zeta_0$, a particular solution $C_P(t)$ is given by

$$C_P(t) = e^{-\zeta\omega t}(P_1 \cos\omega_d t + P_2 \sin\omega_d t), \tag{33}$$

where

$$P_1 = -\frac{r\sqrt{1-\zeta^2}\cdot A}{\omega_0^2\omega_d\cdot\Delta}, \quad P_2 = \frac{(1-\zeta^2)\cdot B}{\omega_0^2\omega_d\cdot\Delta}, \quad r = \frac{\omega}{\omega_0}, \tag{34}$$

and $A = 2(\zeta_0 - \zeta r)$, $B = 1 - r^2 - \zeta r\cdot A$, $\Delta = r^2\cdot A + (1-\zeta^2)\cdot B^2$. For zero initial conditions $y(0) = 0$ and $\dot{y}(0) = 0$, the coefficients C_1 and C_2 of the complementary solution are given by

$$C_1 = -P_1, \quad C_2 = -\frac{A\cdot P_1}{2\sqrt{1-\zeta_0^2}} - \frac{r\sqrt{1-\zeta^2}\cdot P_2}{\sqrt{1-\zeta_0^2}}. \tag{35}$$

Having obtained $C(t) = h_0(t) * h(t)$, the maximum absolute acceleration of the oscillator given by Eq. (29) is

$$|\ddot{u}_F(t)|_{\max} = |(C_1\omega_0^2\omega^2 e^{-\zeta_0\omega_0 t}\cos\omega_{0,d}t + C_2\omega_0^2\omega^2 e^{-\zeta_0\omega_0 t}\sin\omega_{0,d}t + P_1\omega_0^2\omega^2 e^{-\zeta\omega t}\cos\omega_d t + P_2\omega_0^2\omega^2 e^{-\zeta\omega t}\sin\omega_d t) * \ddot{u}_g(t)|_{\max}. \tag{36}$$

Floor response spectra. For lightly damped systems (say, ζ, $\zeta_0 < 0.1$), $\omega_{0,d} \approx \omega_0$ and $\omega_d \approx \omega$. Eq. (36) reduces to

$$\begin{aligned}|\ddot{u}_F(t)|_{\max} &= |(C_1\omega_0^2\omega^2 e^{-\zeta_0\omega_0 t}\cos\omega_0 t + C_2\omega_0^2\omega^2 e^{-\zeta_0\omega_0 t}\sin\omega_0 t + P_1\omega_0^2\omega^2 e^{-\zeta\omega t}\cos\omega t + P_2\omega_0^2\omega^2 e^{-\zeta\omega t}\sin\omega t) * \ddot{u}_g(t)|_{\max}\\ &= |C_1\omega_0\omega^2\cdot[\omega_0\dot{h}_0(t) * \ddot{u}_g(t)] + C_2\omega_0\omega^2\cdot[\omega_0^2 h_0(t) * \ddot{u}_g(t)] + P_1\omega_0^2\omega\cdot[\omega\dot{h}(t) * \ddot{u}_g(t)] + P_2\omega_0^2\omega\cdot[\omega^2 h(t) * \ddot{u}_g(t)]|_{\max}.\end{aligned} \tag{37}$$

The maximum response $|\ddot{u}_F(t)|_{\max}$ may be over-estimated if it is calculated by the sum of the maximum values of each term in Eq. (37), since the maximum values of $|\omega^2 h(t) * \ddot{u}_g(t)|_{\max}$ and $|\omega\dot{h}(t) * \ddot{u}_g(t)|_{\max}$, or $|\omega_0^2 h_0(t) * \ddot{u}_g(t)|_{\max}$ and $|\omega_0\dot{h}_0(t) * \ddot{u}_g(t)|_{\max}$, do not occur simultaneously.

Since there is a $\pi/2$ phase difference between the sine and cosine functions, it is appropriate to employ the SRSS combination rule to calculate the maximum absolute acceleration. Therefore, in non-tuning cases (when the frequencies of the structure and equipment are well separated), the FRS is obtained from Eq. (37) as

$$\mathcal{S}_F^2(\omega_0, \zeta_0) = AF_0^2\cdot\mathcal{S}_A^2(\omega_0, \zeta_0) + AF^2\cdot\mathcal{S}_A^2(\omega, \zeta), \tag{38}$$

in which $\mathcal{S}_F(\omega_0, \zeta_0) = |\ddot{u}_F(t)|_{\max}$ is the FRS or the spectral acceleration of a SDOF oscillator (with circular frequency ω_0 and damping ratio ζ_0) mounted on the SDOF structure (with circular frequency ω and damping ratio ζ), $\mathcal{S}_A(\omega_0, \zeta_0)$ is the GRS or the spectral acceleration of the oscillator mounted on the ground, and AF and AF_0 are the amplification factors discussed in the following subsection.

3.3. Amplification factors

For non-tuning cases, the amplification factors AF_0 and AF are given by

$$\begin{aligned}AF_0 &= \frac{r^2}{\sqrt{(1-r^2)^2 + 4(\zeta_0^2+\zeta^2)r^2 - 4\zeta_0\zeta r(1+r^2)}},\\ AF &= \frac{1}{\sqrt{(1-r^2)^2 + 4(\zeta_0^2+\zeta^2)r^2 - 4\zeta_0\zeta r(1+r^2)}}.\end{aligned} \tag{39}$$

If damping is light and the effect of damping is neglected, the amplification factors are approximately

$$AF_0 \approx \frac{r^2}{1-r^2}, \quad AF \approx \frac{1}{1-r^2}, \quad r = \frac{\omega}{\omega_0}. \tag{40}$$

From Eq. (38), the FRS $\mathcal{S}_F(\omega_0, \zeta_0)$ can be interpreted as a combination of

- amplified spectral acceleration $AF_0\cdot\mathcal{S}_A(\omega_0, \zeta_0)$ of the oscillator, and
- amplified spectral acceleration $AF\cdot\mathcal{S}_A(\omega, \zeta)$ of the structure.

To illustrate the physical meaning of Eq. (38), consider two extreme cases (Fig. 4):

- **Frequency ratio** $r \to \infty$ ($\omega \gg \omega_0$): The structure is very stiff comparing to the oscillator, so that the structure and the ground can be considered as an integral rigid body. The frequency components of ground motions, to which the oscillator is sensitive, are transmitted by the structure without modification. Therefore, the equipment behaves as if it is directly mounted on the ground. When $r \to \infty$, the amplification factors $AF_0 = 1$ and $AF = 0$ agree with this case.
- **Frequency ratio** $r \to 0$ ($\omega \ll \omega_0$): The oscillator is very stiff comparing to the structure or the structure is very flexible comparing to the oscillator, so that the response of the oscillator is the same as that of the structure. When $r = 0$, the amplification factors $AF_0 = 0$ and $AF = 1$, and the maximum response of the oscillator is equal to the spectral acceleration of the structure.

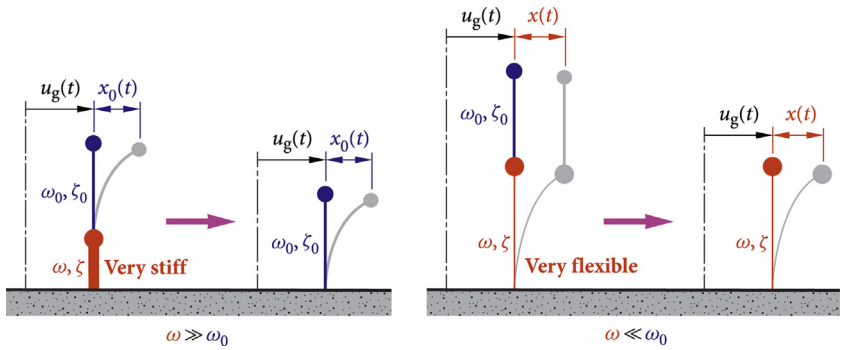
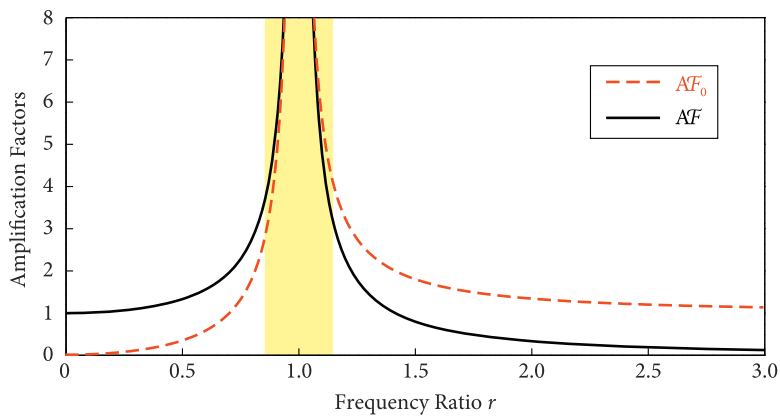
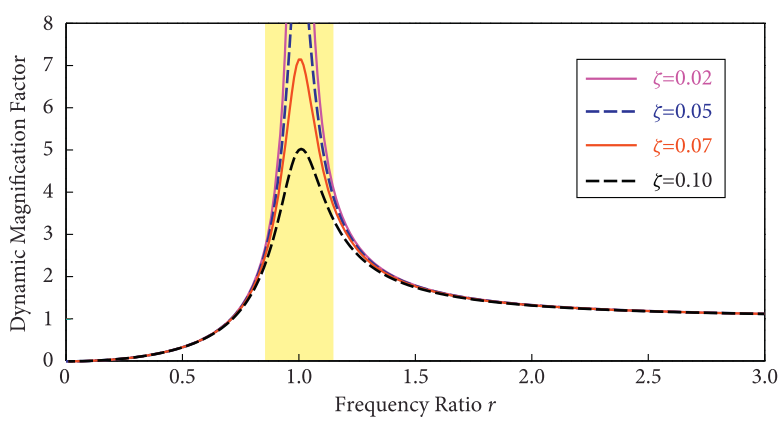


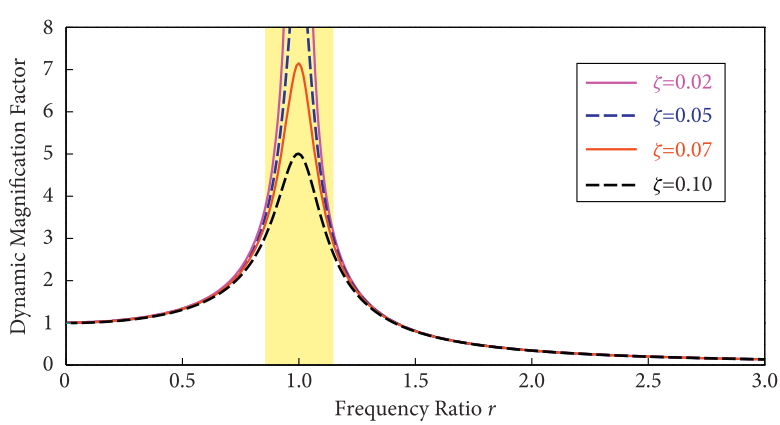
Fig. 4. Two extreme cases of motion amplification.



(a) Amplification Factors of FRS



(b) Dynamic Magnification Factor DMF_0



(c) Dynamic Magnification Factor DMF

Fig. 5. Amplification factors.

The amplification factors AF_0 and AF given by Eqs. (39) and (40) are for non-tuning cases. To extend the concept of amplification factors to perfect-tuning and near-tuning cases, the behavior of the amplification factors given by Eq. (40) is investigated by plotting them in Fig. 5(a):

- The amplification factor of ground motion AF_0 is similar to the Dynamic Magnification Factors (DMF) of a SDOF oscillator subjected to a harmonic loading

$$\mathrm{DMF}_0 = \frac{r^2}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}}. \tag{41}$$

- The amplification factor of structure motion AF is similar to the DMF of a SDOF oscillator under harmonic base excitation

$$\mathrm{DMF} = \frac{1}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}}. \tag{42}$$

DMF_0 and DMF given by Eqs. (41) and (42) are shown in Fig. 5(b) and (c). It is seen that damping has little effect on the response amplification in non-resonant or non-tuning cases (when r is not close to 1), but has a significant effect on the response in perfect-tuning or near-tuning cases (when r approaches 1).

Based on the expressions of DMF_0 and DMF given by Eqs. (41) and (42), when the effect of damping is considered, it is appropriate to assume that the amplification factors AF_0 and AF are of the form, for both tuning and non-tuning cases,

$$AF_0 = \frac{r^2}{\sqrt{(1-r^2)^2 + (2\zeta_{0,\mathrm{e}} r)^2}}, \qquad AF = \frac{1}{\sqrt{(1-r^2)^2 + (2\zeta_{\mathrm{e}} r)^2}}, \tag{43}$$

in which $\zeta_{0,\mathrm{e}}$ and ζ_{e} are considered to be the *equivalent damping coefficients* for the amplification factors of ground motion and structure motion, respectively.

In the non-tuning cases, the amplification factors given by Eqs. (39) and (40) can be used directly; it is not necessary to specify the equivalent damping coefficients $\zeta_{0,\mathrm{e}}$ and ζ_{e} in Eq. (43).

In the following subsection, the tuning case is investigated to quantify the equivalent damping coefficients $\zeta_{0,\mathrm{e}}$ and ζ_{e}.

3.4. Perfect-tuning case

When $\omega_0 = \omega$ and for small damping ζ_0, $\zeta < 0.1$, $C(t) = h_0(t) * h(t)$ becomes

$$h_0(t) * h(t) = \int_0^t \frac{1}{\omega_0} \mathrm{e}^{-\zeta_0\omega_0(t-\tau)} \sin\omega_0(t-\tau) \cdot \frac{1}{\omega} \mathrm{e}^{-\zeta\omega\tau} \sin\omega\tau \mathrm{d}\tau = \frac{1}{\omega^3[4+(\zeta-\zeta_0)^2]}$$
$$\left[\frac{2}{\zeta-\zeta_0}(\mathrm{e}^{-\zeta\omega t} - \mathrm{e}^{-\zeta_0\omega t})\cos\omega t + (\mathrm{e}^{-\zeta\omega t} + \mathrm{e}^{-\zeta_0\omega t})\sin\omega t\right],$$

which can be simplified to, for small damping $(\zeta - \zeta_0) \to 0$,

$$h_0(t) * h(t) = \frac{1}{2\omega^3(\zeta-\zeta_0)}(\mathrm{e}^{-\zeta\omega t} - \mathrm{e}^{-\zeta_0\omega t})\cos\omega t + \frac{1}{4\omega^3}(\mathrm{e}^{-\zeta\omega t} + \mathrm{e}^{-\zeta_0\omega t})\sin\omega t = \frac{\dot{h}(t) - \dot{h}_0(t)}{2\omega^3(\zeta-\zeta_0)} + \frac{h(t)+h_0(t)}{4\omega^2}. \tag{44}$$

Substituting Eq. (44) into (29) yields the maximum response of the oscillator

$$|\ddot{u}_{\mathrm{F}}(t)|_{\max} = \left|\frac{\omega\dot{h}(t) * \ddot{u}_{\mathrm{g}}(t) - \omega\dot{h}_0(t) * \ddot{u}_{\mathrm{g}}(t)}{2(\zeta-\zeta_0)} + \frac{\omega^2 h(t) * \ddot{u}_{\mathrm{g}}(t) + \omega^2 h_0(t) * \ddot{u}_{\mathrm{g}}(t)}{4}\right|_{\max} = \left|\frac{1}{2} \cdot \frac{\ddot{u}(t) - \ddot{u}_0(t)}{\zeta-\zeta_0} + \frac{\ddot{u}(t) + \ddot{u}_0(t)}{4}\right|_{\max}, \tag{45}$$

in which the following relationships have been used

$$u(t) = h(t) * \ddot{u}_{\mathrm{g}}(t), \quad \dot{u}(t) = \dot{h}(t) * \ddot{u}_{\mathrm{g}}(t), \quad \ddot{u}(t) = \omega^2 h(t) * \ddot{u}_{\mathrm{g}}(t) = \omega\dot{h}(t) * \ddot{u}_{\mathrm{g}}(t).$$

When $\zeta_0 = \zeta$, $\ddot{u}(t) = \ddot{u}_0(t)$; the first term in Eq. (45), which is dominant, is undefined. For $(\zeta - \zeta_0) \to 0$, Eq. (45) becomes

$$|\ddot{u}_{\mathrm{F}}(t)|_{\max} = \frac{1}{2}\left|\frac{\partial\ddot{u}(t)}{\partial\zeta} + \ddot{u}(t)\right|_{\max}. \tag{46}$$

Differentiating $\ddot{u}(t) = \omega\dot{h}(t) * \ddot{u}_{\mathrm{g}}(t)$ with respect to ζ gives

$$\frac{\partial\ddot{u}(t)}{\partial\zeta} = \frac{\partial[\omega\dot{h}(t) * \ddot{u}_{\mathrm{g}}(t)]}{\partial\zeta} = -\omega^2 t\mathrm{e}^{-\zeta\omega t}\cos\omega t * \ddot{u}_{\mathrm{g}}(t). \tag{47}$$

Note that $\ddot{u}(t)$ can also be written as $\ddot{u}(t) = \omega^2 h(t) * \ddot{u}_{\mathrm{g}}(t) = \omega\mathrm{e}^{-\zeta\omega t}\sin\omega t * \ddot{u}_{\mathrm{g}}(t)$.

Hence, in the perfect-tuning case with $\omega_0 = \omega$, $\zeta_0 = \zeta$, FRS given by Eq. (46) becomes

$$\mathcal{S}_{\mathrm{F}}(\omega,\zeta) = \frac{1}{2}\left|-\omega^2 t\mathrm{e}^{-\zeta\omega t}\cos\omega t * \ddot{u}_{\mathrm{g}}(t) + \omega\mathrm{e}^{-\zeta\omega t}\sin\omega t * \ddot{u}_{\mathrm{g}}(t)\right|_{\max} = \mathcal{S}_{\mathrm{A}}^{\mathrm{t}}(\omega,\zeta). \tag{48}$$

Analogous to GRS defined in (30), Eq. (48) is defined as t-Response Spectrum (tRS), in which "t" indicates "tuning" or the extra "t" variable in the first convolution term as compared to GRS defined in Eq. (30). The concepts of GRS and tRS are illustrated in Fig. 6. Under an earthquake excitation $\ddot{u}_{\mathrm{g}}(t)$,

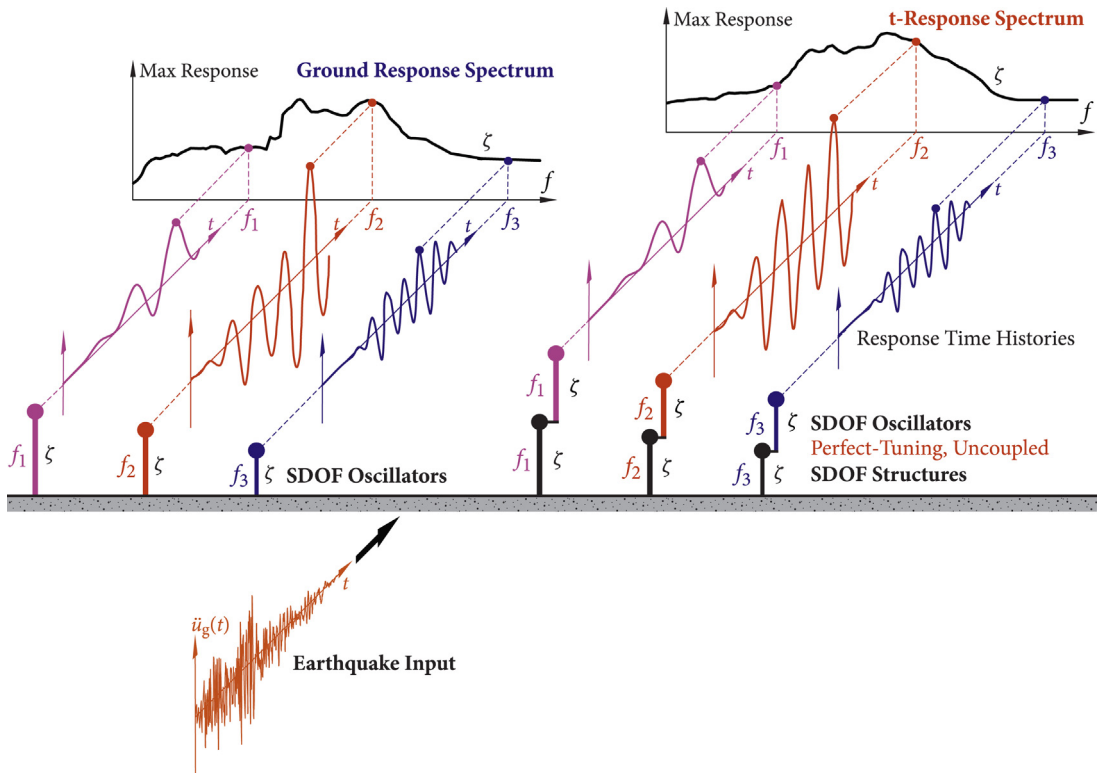


Fig. 6. Seismic analysis methods for secondary systems.

- GRS $\mathcal{S}_A(f, \zeta)$ is the maximum acceleration response of a SDOF oscillator (with frequency f and damping ratio ζ) mounted directly on ground;
- tRS $\mathcal{S}_A^t(f, \zeta)$ is the maximum acceleration response of a SDOF oscillator (with frequency f and damping ratio ζ) mounted on top of a SDOF structure (with the same f and ζ) that is mounted on ground. The identical SDOF oscillator and SDOF structure are uncoupled and are in *resonance* or *tuning*.

FRS given by Eq. (48) can be expressed in the form of Eq. (38). Note that, in the perfect-tuning case, $\omega_0 = \omega$, $r = 1$, $\zeta_0 = \zeta$, and $\mathcal{S}_A(\omega_0, \zeta_0) = \mathcal{S}_A(\omega, \zeta)$, $AF_0 = AF$. Eq. (38) reduces to

$$\mathcal{S}_F(\omega_0, \zeta_0) = \sqrt{2} \cdot AF_0 \cdot \mathcal{S}_A(\omega_0, \zeta_0) = \mathcal{S}_A^t(\omega_0, \zeta_0), \tag{49}$$

which gives

$$AF_0 = AF = \frac{1}{\sqrt{2}} \cdot \frac{\mathcal{S}_A^t(\omega_0, \zeta_0)}{\mathcal{S}_A(\omega_0, \zeta_0)}. \tag{50}$$

From Eq. (43), when $r = 1$, one has

$$AF_0 = \frac{r^2}{\sqrt{(1-r^2)^2 + (2\zeta_{0,e} r)^2}}\Big|_{r=1} = \frac{r^2}{2\zeta_{0,e}}, \qquad AF = \frac{1}{2\zeta_e}. \tag{51}$$

Hence, from Eqs. (50) and (51), the equivalent damping coefficients for the perfect-tuning case are given by

$$\zeta_{0,e} = \zeta_e = \frac{1}{\sqrt{2}} \cdot \frac{\mathcal{S}_A(\omega_0, \zeta_0)}{\mathcal{S}_A^t(\omega_0, \zeta_0)}. \tag{52}$$

Near-tuning cases. In near-tuning cases, there is no exact formulation to determine the equivalent damping coefficients $\zeta_{0,e}$ and ζ_e. The following approximations may be used:

- $r \approx 1$, $\zeta_0 = \zeta$: The equivalent damping coefficients for the perfect-tuning case given by Eq. (52) can be used.
- $r \not\approx 1$, $\zeta_0 \neq \zeta$: The following approximations may be used

$$\zeta_{0,e} \approx \frac{\sqrt{2}\zeta_0}{\zeta_0 + \zeta} \cdot \frac{\mathcal{S}_A(\omega_0, \zeta_0)}{\mathcal{S}_A^t(\omega_0, \zeta_0)}, \qquad \zeta_e \approx \frac{\sqrt{2}\zeta}{\zeta_0 + \zeta} \cdot \frac{\mathcal{S}_A(\omega, \zeta)}{\mathcal{S}_A^t(\omega, \zeta)}. \tag{53}$$

The appropriateness of these approximations is verified by numerical simulations presented in Section 4.

Note that, when $\zeta_0 \to \zeta$, Eq. (53) reduces to Eq. (52). Hence, combining the tuning and near-tuning cases, the equivalent damping coefficients can be taken as, for all cases,

$$\zeta_{0,\mathrm{e}} = \frac{\sqrt{2}\zeta_0}{\zeta_0+\zeta}\cdot\frac{\mathcal{S}_{\mathrm{A}}(\omega_0,\zeta_0)}{\mathcal{S}^{\mathrm{t}}_{\mathrm{A}}(\omega_0,\zeta_0)}, \qquad \zeta_{\mathrm{e}} = \frac{\sqrt{2}\zeta}{\zeta_0+\zeta}\cdot\frac{\mathcal{S}_{\mathrm{A}}(\omega,\zeta)}{\mathcal{S}^{\mathrm{t}}_{\mathrm{A}}(\omega,\zeta)}. \tag{54}$$

Comments on approximations of FRS. In Eq. (48), FRS is equal to tRS for the perfect-tuning case. However, due to the presence of a time variable t in the first convolution term, it is difficult to obtain an analytical expression for Eq. (48) in terms of GRS.

Over the past decades, researchers (Yasui et al., 1993; An et al., 2013) have made some approximations to evaluate Eq. (45).

The second term on the right-hand side of Eq. (45) may be neglected, since its contribution is far less than that of the dominant first term. Therefore, Eq. (45) becomes

$$|\ddot{u}_{\mathrm{F}}(t)|_{\max} \approx |\frac{\ddot{u}(t)-\ddot{u}_0(t)}{2(\zeta-\zeta_0)}|_{\max}. \tag{55}$$

A similar formula is derived in Yasui et al. (1993) through integration by parts

$$2\zeta\ddot{u}_{\mathrm{F}}(t) - 2\zeta_0\ddot{u}_{\mathrm{F}}(t+\Delta t) \approx \ddot{u}(t) - \ddot{u}_0(t). \tag{56}$$

It is assumed that the two terms on the left-hand side of Eq. (56) have different phase angles, and the maximum value of the left-hand side is obtained by summing the maximum absolute value of each term

$$2(\zeta+\zeta_0)|\ddot{u}_{\mathrm{F}}(t)|_{\max} = |\ddot{u}(t)-\ddot{u}_0(t)|_{\max}. \tag{57}$$

Since the maximum values $|\ddot{u}(t)|_{\max} = \mathcal{S}_{\mathrm{A}}(\omega,\zeta)$ and $|\ddot{u}_0(t)|_{\max} = \mathcal{S}_{\mathrm{A}}(\omega_0,\zeta_0)$ do not occur simultaneously, the SRSS combination rule is applied to evaluate the right-hand side of Eq. (57). Hence, the maximum response of the oscillator or FRS is given by

$$\mathcal{S}_{\mathrm{F}}(\omega_0,\zeta_0) = |\ddot{u}_{\mathrm{F}}(t)|_{\max} = \frac{\sqrt{\mathcal{S}^2_{\mathrm{A}}(\omega,\zeta)+\mathcal{S}^2_{\mathrm{A}}(\omega_0,\zeta_0)}}{2(\zeta+\zeta_0)}. \tag{58}$$

In the perfect-tuning case, $\omega_0 = \omega$, $\zeta_0 = \zeta$, $\mathcal{S}_{\mathrm{A}}(\omega_0,\zeta_0) = \mathcal{S}_{\mathrm{A}}(\omega,\zeta)$, it reduces to

$$\mathcal{S}_{\mathrm{F}}(\omega_0,\zeta_0) = \frac{1}{2\sqrt{2}\zeta_0}\cdot\mathcal{S}_{\mathrm{A}}(\omega_0,\zeta_0). \tag{59}$$

However, these approximations are reasonable only when the phase differences between the two terms on the left-hand side and right-hand side of Eq. (57) are π and $\pi/2$, respectively. In this case, the maximum values of the two sides can be obtained by the sum of absolute values and the SRSS rules, respectively.

Otherwise, this assumption, which gives a constant amplification factor $1/(2\sqrt{2}\zeta_0)$, can be inaccurate. For example, considering an extreme case for $f_0 = f = 100$ Hz, the structure and equipment are both extremely stiff so that the maximum acceleration of the equipment should be close to the PGA of GRS, i.e., $\mathcal{S}_{\mathrm{F}}(\omega_0,\zeta_0) = \mathcal{S}_{\mathrm{A}}(\omega_0,\zeta_0)$. For a given damping coefficient ζ_0, the amplification factor for the perfect-tuning case should be dependent on the frequency instead of being a constant.

To obtained an accurate estimation of tRS, statistical relationships between tRS and GRS are established from simulation results using a large number of real earthquake records in the companion paper.

3.5. SDOF oscillator mounted on multiple DOF structure

Since almost all engineering structures have multiple DOF, the formulation for an oscillator mounted on a SDOF structure is extended to an oscillator mounted on a multiple DOF structure in this subsection.

Consider a three-dimensional model of a structure with N nodes, in which each node has six DOF, subjected to tri-directional earthquake excitations, as shown in Fig. 2. The equation of motion in the matrix form is given by Eq. (6).

Applying modal analysis as presented in Section 2.3, the 6N-DOF system (7) is reduced to a series of 6N SDOF systems, in which the modal displacement $q^i_k(t)$ of the kth mode (SDOF system) under the earthquake excitation in direction i is governed by Eq. (15). The absolute acceleration $\ddot{u}^i_{n,j}(t)$ of node N in direction j under the earthquake excitation in direction i is given by Eq. (16), which is a linear combination of all 6N modal responses and the contribution factor of the kth modal response q^i_k is $\varphi_{n,j;k}\Gamma^i_k$.

Maximum modal response contribution. As derived in Section 3.3, for a SDOF oscillator mounted on a SDOF structure, the maximum response of the oscillator is given by Eq. (38). Therefore, from Eqs. (15)–(19), for the maximum absolute acceleration $\mathcal{S}^i_{n,j}(\omega_0,\zeta_0)$ in direction j of an oscillator (with frequency ω_0 and damping ratio ζ_0) mounted at node N under the earthquake excitation in direction i, the maximum contribution by the kth mode, i.e., the maximum absolute acceleration of the oscillator under the excitation $\ddot{u}^i_{n,j;k}$ of Eq. (16), is given by

$$R^i_{n,j;k} = \varphi_{n,j;k}\Gamma^i_k\sqrt{\{\mathrm{AF}^2_{0,k}\cdot[\mathcal{S}^i_{\mathrm{A}}(\omega_0,\zeta_0)]^2 + \mathrm{AF}^2_k\cdot[\mathcal{S}^i_{\mathrm{A}}(\omega_k,\zeta_k)]^2\}}, \tag{60}$$

where $\mathcal{S}^i_{\mathrm{A}}(\omega,\zeta)$ is GRS of the earthquake excitation in direction i, and $\mathrm{AF}_{0,k}$ and AF_k are the amplification factors of the kth mode given by

$$\mathrm{AF}_{0,k} = \frac{r^2_k}{\sqrt{(1-r^2_k)^2+(2\zeta_{0,\mathrm{e}}r_k)^2}}, \quad \mathrm{AF}_k = \frac{1}{\sqrt{(1-r^2_k)^2+(2\zeta_{k,\mathrm{e}}r_k)^2}}, \quad r_k = \frac{\omega_k}{\omega_0}. \tag{61}$$

Modal combination: FRS-CQC. Since the maximum responses $R^i_{n,j;k}$ of the oscillator contributed to $\mathcal{S}^i_{n,j}(\omega_0,\zeta_0)$ by each of the k modes ($k = 1, 2, \ldots, 6N$) do no occur simultaneously, they have to be combined following an appropriate combination rule.

Comparing Eqs. (2)–(5) with Eqs. (15)–(18), and using Eq. (29), the contribution from the kth mode to the response of the oscillator (with frequency ω_0 and damping ratio ζ_0) mounted on the multiple DOF structure under the earthquake excitation in direction i is approximately given by

$$Q_k^i(t) = \omega_0^2\omega_k^2 \cdot h_0(t) * h_k(t) * \ddot{u}_g^i(t) = \omega_0^2\omega_k^2 \cdot \mathsf{C}_k(t) * \ddot{u}_g^i(t), \tag{62}$$

where $\mathsf{C}_k(t) = \mathsf{h}_0(t) * \mathsf{h}_k(t)$. The covariance between $q_k^i(t)$ and $Q_\kappa^i(t)$ of modes k and κ is given by

$$\begin{aligned}\mathsf{E}[Q_k^i(t)Q_\kappa^i(t+\tau)] &= \omega_0^4\omega_k^2\omega_\kappa^2 \cdot \mathsf{E}\left[\int_{-\infty}^{\infty} \mathsf{C}_k(\tau_1)\,\ddot{u}_g^i(t-\tau_1)\,\mathrm{d}\tau_1 \int_{-\infty}^{\infty} \mathsf{C}_\kappa(\tau_2)\,\ddot{u}_g^i(t+\tau-\tau_2)\,\mathrm{d}\tau_2\right] = \omega_0^4\omega_k^2\omega_\kappa^2 \int_{-\infty}^{\infty} \\ &\int_{-\infty}^{\infty} \mathsf{C}_k(\tau_1)\mathsf{C}_\kappa(\tau_2)\,\mathsf{E}[\ddot{u}_g^i(t-\tau_1)\,\ddot{u}_g^i(t+\tau-\tau_2)]\mathrm{d}\tau_1\mathrm{d}\tau_2 = \omega_0^4\omega_k^2\omega_\kappa^2 \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \mathsf{C}_k(\tau_1)\mathsf{C}_\kappa(\tau_2) R_{\ddot{u}_g^i\ddot{u}_g^i}(\tau+\tau_1-\tau_2)\,\mathrm{d}\tau_1\mathrm{d}\tau_2.\end{aligned} \tag{63}$$

Taking Fourier transform of both sides yields

$$\mathsf{S}_{Q_k^iQ_\kappa^i}(\omega) = \int_{-\infty}^{\infty} \mathsf{E}[Q_k^i(t)Q_\kappa^i(t+\tau)] \cdot \mathrm{e}^{-\mathrm{i}\omega\tau}\mathrm{d}\tau = \omega_0^4\omega_k^2\omega_\kappa^2 \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \mathsf{C}_k(\tau_1)\mathsf{C}_\kappa(\tau_2) R_{\ddot{u}_g^i\ddot{u}_g^i}(\tau+\tau_1-\tau_2)\cdot \mathrm{e}^{-\mathrm{i}\omega\tau}\mathrm{d}\tau_1\mathrm{d}\tau_2\mathrm{d}\tau. \tag{64}$$

Setting $\tau_3 = \tau + \tau_1 - \tau_2$, Eq. (62) can be written as

$$\mathsf{S}_{Q_k^iQ_\kappa^i}(\omega) = \omega_0^4\omega_k^2\omega_\kappa^2 \int_{-\infty}^{\infty} \mathsf{C}_k(\tau_1)\mathrm{e}^{\mathrm{i}\omega\tau_1}\mathrm{d}\tau_1 \int_{-\infty}^{\infty} \mathsf{C}_\kappa(\tau_2)\mathrm{e}^{-\mathrm{i}\omega\tau_2}\mathrm{d}\tau_2 \int_{-\infty}^{\infty} R_{\ddot{u}_g^i\ddot{u}_g^i}(\tau_3)\mathrm{e}^{-\mathrm{i}\omega\tau_3}\mathrm{d}\tau = \omega_k^2\omega_\kappa^2 \cdot \mathcal{C}_k{}^*(\omega)\cdot\mathcal{C}_\kappa(\omega)\cdot \mathsf{S}_{\ddot{u}_g^i\ddot{u}_g^i}(\omega), \tag{65}$$

where $\mathcal{C}_k(\omega) = H_0(\omega)H_k(\omega)$ is the Fourier transform of $\mathsf{C}_k(t) = \mathsf{h}_0(t) * \mathsf{h}_k(t)$, $\mathcal{C}_k{}^*(\omega)$ is the complex conjugate of $\mathcal{C}_k(\omega)$, and $\mathsf{S}_{\ddot{u}_g^i\ddot{u}_g^i}(\omega)$ is the power spectral density of the earthquake excitation $\ddot{u}_g^i(t)$.

Taking the inverse Fourier transform of Eq. (65) yields

$$\mathsf{E}[Q_k^i(t)Q_\kappa^i(t+\tau)] = \frac{1}{2\pi}\int_{-\infty}^{\infty} \mathsf{S}_{Q_k^iQ_\kappa^i}(\omega)\mathrm{e}^{\mathrm{i}\omega\tau}\mathrm{d}\omega = \frac{\omega_0^4\omega_k^2\omega_\kappa^2}{2\pi}\int_{-\infty}^{\infty} \mathcal{C}_k{}^*(\omega)\cdot\mathcal{C}_\kappa(\omega)\cdot \mathsf{S}_{\ddot{u}_g^i\ddot{u}_g^i}(\omega)\mathrm{e}^{\mathrm{i}\omega\tau}\mathrm{d}\omega. \tag{66}$$

Setting $\tau = 0$ results in

$$\mathsf{E}[Q_k^i(t)Q_\kappa^i(t)] = \frac{\omega_0^4\omega_k^2\omega_\kappa^2}{2\pi}\int_{-\infty}^{\infty} \mathcal{C}_k{}^*(\omega)\cdot\mathcal{C}_\kappa(\omega)\cdot \mathsf{S}_{\ddot{u}_g^i\ddot{u}_g^i}(\omega)\mathrm{d}\omega = \frac{\omega_0^4\omega_k^2\omega_\kappa^2}{2\pi}\int_{-\infty}^{\infty} H_0^*(\omega)H_k^*(\omega)\cdot H_0(\omega)H_\kappa(\omega)\cdot \mathsf{S}_{\ddot{u}_g^i\ddot{u}_g^i}(\omega)\mathrm{d}\omega. \tag{67}$$

Since ground motions can be generally modelled as wide-band noises, it is reasonable to assume the seismic input $\ddot{u}_g^i(t)$ as a white noise by letting the power spectral density $\mathsf{S}_{\ddot{u}_g^i\ddot{u}_g^i}(\omega) = \mathsf{S}^i$. Therefore, Eq. (67) can be written as

$$\mathsf{E}[Q_k^i(t)Q_\kappa^i(t)] = \frac{\omega_0^4\omega_k^2\omega_\kappa^2}{2\pi}\cdot \mathsf{S}^i \cdot I_{k\kappa}, \tag{68}$$

where

$$\begin{aligned} I_{k\kappa} &= \int_{-\infty}^{\infty} H_0^*(\omega)H_k^*(\omega)\cdot H_0(\omega)H_\kappa(\omega)\mathrm{d}\omega = \int_{-\infty}^{\infty} \frac{1}{(\omega_0^2-\omega^2)^2+(2\zeta_0\omega_0\omega)^2}\cdot\frac{1}{(\omega_k^2-\omega^2)-\mathrm{i}2\zeta_k\omega_k\omega}\cdot\frac{1}{(\omega_\kappa^2-\omega^2)+\mathrm{i}2\zeta_\kappa\omega_\kappa\omega}\mathrm{d}\omega \\ &= \int_{-\infty}^{\infty} \frac{[(\omega_k^2-\omega^2)+\mathrm{i}2\zeta_k\omega_k\omega]\cdot[(\omega_\kappa^2-\omega^2)-\mathrm{i}2\zeta_\kappa\omega_\kappa\omega]}{\Delta_{k\kappa}}\mathrm{d}\omega = \Re(I_{k\kappa}) + \mathrm{i}\Im(I_{k\kappa}) \end{aligned} \tag{69}$$

where

$$\Delta_{k\kappa} = [(\omega_0^2-\omega^2)^2+(2\zeta_0\omega_0\omega)^2]\cdot[(\omega_k^2-\omega^2)^2+(2\zeta_k\omega_k\omega)^2]\cdot[(\omega_\kappa^2-\omega^2)^2+(2\zeta_\kappa\omega_\kappa\omega)^2],$$

$\Re(I_{k\kappa})$ and $\Im(I_{k\kappa})$ are the real and imaginary parts of $I_{k\kappa}$, respectively, given by

$$\begin{aligned}\Re(I_{k\kappa}) &= \int_{-\infty}^{\infty} \frac{(\omega_k^2-\omega^2)\cdot(\omega_\kappa^2-\omega^2)+(2\zeta_k\omega_k\omega)\cdot(2\zeta_\kappa\omega_\kappa\omega)}{\Delta_{k\kappa}}\mathrm{d}\omega, \\ \Im(I_{k\kappa}) &= \int_{-\infty}^{\infty} \frac{2\zeta_k\omega_k\omega\cdot(\omega_\kappa^2-\omega^2)-2\zeta_\kappa\omega_\kappa\omega\cdot(\omega_k^2-\omega^2)}{\Delta_{k\kappa}}\mathrm{d}\omega,\end{aligned}$$

which can be evaluated by the method of residue to yield

$$\Re(I_{k\kappa}) = \frac{\pi\cdot\alpha_{k\kappa}}{2\zeta_0\omega_0^3\cdot\omega_0^4}, \qquad \Im(I_{k\kappa}) = 0, \tag{70}$$

in which

$$\alpha_{k\kappa} = \prod_{l=1}^{3} D_{k\kappa,l}^{-1}\cdot\sum_{l=0}^{4} C_{k\kappa,l}\,\zeta_0^l, \tag{71}$$

$C_{k\kappa,l}$ and $D_{k\kappa,l}$ are constants in terms of ζ_0, ζ_k, ζ_κ, $r_k = \omega_k/\omega_0$, and $r_\kappa = \omega_\kappa/\omega_0$, given by

$$D_{k\kappa,1} = 1 - 2r_\kappa^2 + r_\kappa^4 + 4\zeta_0\zeta_\kappa r_\kappa + 4\zeta_0\zeta_\kappa r_\kappa^3 + 4\zeta_0^2 r_\kappa^2 + 4\zeta_\kappa^2 r_\kappa^2,$$
$$D_{k\kappa,2} = 1 - 2r_k^2 + r_k^4 + 4\zeta_0\zeta_k r_k + 4\zeta_0\zeta_k r_k^3 + 4\zeta_0^2 r_k^2 + 4\zeta_k^2 r_k^2,$$
$$D_{k\kappa,3} = (r_k^2 - r_\kappa^2)^2 + 4\zeta_k\zeta_\kappa r_k r_\kappa (r_k^2 + r_\kappa^2) + 4r_k^2 r_\kappa^2(\zeta_k^2 + \zeta_\kappa^2),$$
$$C_{k\kappa,0} = (1 - r_k^2 - r_\kappa^2 + r_k^2 r_\kappa^2 + 4\zeta_k\zeta_\kappa r_k r_\kappa)\cdot D_{k\kappa,3},$$

$$\begin{aligned} C_{k\kappa,1}/4 = {} & 2\zeta_k r_k + 2\zeta_\kappa r_\kappa + 8\zeta_k\zeta_\kappa r_k r_\kappa(\zeta_k r_k + \zeta_\kappa r_\kappa) - 4\zeta_k r_k(r_k^2 + r_\kappa^2) + 8\zeta_k^3 r_k^3 + 8\zeta_\kappa^3 r_\kappa^3 - 2r_k r_\kappa(\zeta_k r_\kappa^3 + \zeta_\kappa r_k^3) + 8\zeta_k\zeta_\kappa r_k r_\kappa(\zeta_k r_k^3 + \zeta_\kappa r_\kappa^3) \\ & + 4r_k^2 r_\kappa^2(\zeta_k r_k + \zeta_\kappa r_\kappa) - 8r_k^2 r_\kappa^2(\zeta_k^3 r_k + \zeta_\kappa^3 r_\kappa) - 8\zeta_k\zeta_\kappa r_k^2 r_\kappa^2(\zeta_k r_\kappa + \zeta_\kappa r_k) + 32\zeta_k^2\zeta_\kappa^2 r_k^2 r_\kappa^2(\zeta_k r_k + \zeta_\kappa r_\kappa) + r_k r_\kappa(\zeta_k r_\kappa^5 + \zeta_\kappa r_k^5) \\ & + r_k^2 r_\kappa^2(\zeta_k r_k^3 + \zeta_\kappa r_\kappa^3) + 4\zeta_k\zeta_\kappa r_k^2 r_\kappa^2(\zeta_k r_\kappa^3 + \zeta_\kappa r_k^3) - 2r_k^3 r_\kappa^3(\zeta_k r_\kappa + \zeta_\kappa r_k) + 4r_k^3 r_\kappa^3(\zeta_k^3 r_\kappa + \zeta_\kappa r_k) + 8\zeta_k\zeta_\kappa r_k^3 r_\kappa^3(\zeta_k r_k + \zeta_\kappa r_\kappa), \end{aligned}$$

$$\begin{aligned} C_{k\kappa,2}/4 = {} & 8\zeta_k^2 r_k^2 + 8\zeta_\kappa^2 r_\kappa^2 + 16\zeta_k\zeta_\kappa r_k r_\kappa + 64\zeta_k^2\zeta_\kappa^2 r_k^2 r_\kappa^2 - 4\zeta_k\zeta_\kappa r_k r_\kappa(r_k^2 + r_\kappa^2) + 32\zeta_k\zeta_\kappa r_k r_\kappa(\zeta_k^2 r_k^2 + \zeta_\kappa^2 r_\kappa^2) + 6r_k^2 r_\kappa^2 - 12r_k^2 r_\kappa^2(\zeta_k^2 + \zeta_\kappa^2) \\ & - 3(r_k^4 + r_\kappa^4) + 8\zeta_k\zeta_\kappa r_k r_\kappa(r_k^4 + r_\kappa^4) - r_k^2 r_\kappa^2(r_k^2 + r_\kappa^2) + 8\zeta_k^2 r_k^4 + 8\zeta_\kappa^2 r_\kappa^4 + 4r_k^2 r_\kappa^2(\zeta_k^2 + \zeta_\kappa^2)(r_k^2 + r_\kappa^2) + 16\zeta_k^2\zeta_\kappa^2 r_k^2 r_\kappa^2(r_k^2 + r_\kappa^2) \\ & + 16\zeta_k\zeta_\kappa r_k^3 r_\kappa^3(\zeta_k^2 + \zeta_\kappa^2) + r_k^6 + r_\kappa^6, \end{aligned}$$

$$\begin{aligned} C_{k\kappa,3}/16 = {} & 8\zeta_k\zeta_\kappa r_k r_\kappa(\zeta_k r_k + \zeta_\kappa r_\kappa) + 2\zeta_k r_k^3 + 2\zeta_\kappa r_\kappa^3 + r_k r_\kappa(\zeta_k r_\kappa^3 + \zeta_\kappa r_k^3) + 4\zeta_k\zeta_\kappa r_k r_\kappa(\zeta_k r_k^3 + \zeta_\kappa r_\kappa^3) - 2r_k^2 r_\kappa^2(\zeta_k r_k + \zeta_\kappa r_\kappa) \\ & + 4r_k^2 r_\kappa^2(\zeta_k^3 r_k + \zeta_\kappa^3 r_\kappa) + 8\zeta_k\zeta_\kappa r_k^2 r_\kappa^2(\zeta_k r_\kappa + \zeta_\kappa r_k) + \zeta_k r_k^5 + \zeta_\kappa r_\kappa^5, \end{aligned}$$

$$C_{k\kappa,4}/16 = D_{k\kappa,3}.$$

Therefore, Eq. (68) can be expressed as

$$\mathsf{E}[Q_k^i(t)Q_\kappa^i(t)] = \frac{\omega_0^4\omega_k^2\omega_\kappa^2}{2\pi}\cdot \mathsf{S}^i \cdot \frac{\pi\cdot\alpha_{k\kappa}}{2\zeta_0\omega_0^3\cdot\omega_0^4} = \frac{\omega_0\mathsf{S}^i}{4\zeta_0}\cdot\alpha_{k\kappa}\cdot r_k^2 r_\kappa^2. \tag{72}$$

When $\kappa = k$, Eq. (72) becomes

$$\mathsf{E}[\{Q_k^i(t)\}^2] = \frac{\omega_0\mathsf{S}^i}{4\zeta_0}\cdot\beta_k\cdot r_k^4, \tag{73}$$

where

$$\beta_k = \frac{\zeta_0 + 4\zeta_0^2\zeta_k r_k + 4\zeta_0\zeta_k^2 r_k^2 + \zeta_k r_k^3}{\zeta_k r_k^3(1 - 2r_k^2 + r_k^4 + 4\zeta_0\zeta_k r_k + 4\zeta_k^2 r_k^2 + 4\zeta_0^2 r_k^2 + 4\zeta_0\zeta_k r_k^3)}. \tag{74}$$

Hence, the correlation coefficient between the contributions to the response of an oscillation mounted on the structure under an earthquake excitation in direction i by kth and κth modes is obtained as

$$\rho_{k\kappa}^i = \frac{\mathsf{E}[Q_k^i(t)Q_\kappa^i(t)]}{\sqrt{\mathsf{E}[\{Q_k^i(t)\}^2]\cdot\mathsf{E}[\{Q_\kappa^i(t)\}^2]}} = \frac{\frac{\omega_0\mathsf{S}^i}{4\zeta_0}\cdot\alpha_{k\kappa}\cdot r_k^2 r_\kappa^2}{\sqrt{\frac{\omega_0\mathsf{S}^i}{4\zeta_0}\cdot\beta_k\cdot r_k^4 \times \frac{\omega_0\mathsf{S}^i}{4\zeta_0}\cdot\beta_\kappa\cdot r_\kappa^4}} = \frac{\alpha_{k\kappa}}{\sqrt{\beta_k\beta_\kappa}},$$

which is independent of the direction i of the earthquake excitation and can be written as

$$\rho_{k\kappa} = \frac{\alpha_{k\kappa}}{\sqrt{\beta_k\beta_\kappa}}. \tag{75}$$

Combining the maximum absolute acceleration $R_{n,j;k}^i$ of the oscillator contributed by mode k, given by Eq. (60), for all $6N$ modes gives the FRS of node N in direction j under the earthquake excitation in direction i defined by Eq. (19):

$$\mathcal{S}_{n,j}^i(\omega_0, \zeta_0) = |\ddot{u}_{\mathrm{F},n,j}^i(t)|_{\max} = \sqrt{\sum_{k=0}^{6N}\sum_{\kappa=0}^{6N}\rho_{k\kappa}R_k^i R_\kappa^i}. \tag{76}$$

FRS $\mathcal{S}_{n,j}(\omega_0, \zeta_0)$ of the Nth node in direction j under tri-directional earthquake excitations is then obtained from FRS $\mathcal{S}_{n,j}^i(\omega_0, \zeta_0)$, $i = 1, 2, 3$, using the SRSS combination rule given by Eq. (20).

Comments on modal combination. Since the modal combination in Eq. (76) is a complete quadrature for maximum responses of the oscillator contributed by all $6N$ modes, it is therefore called FRS-CQC to differentiate it from CQC (Complete Quadratic Combination), which combines maximum responses of the $6N$ modes.

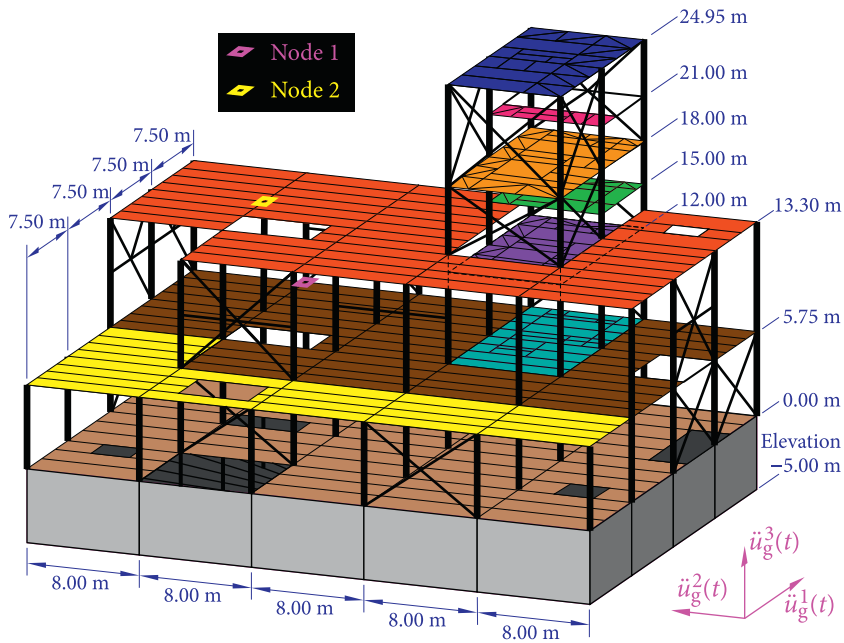


Fig. 7. 3D finite element model of service building.

- To determine responses of multiple DOF structures under earthquake excitations using a response spectrum method, the correlation coefficient between two modal responses is determined for CQC (Der Kiureghian, 1980, 1981), i.e.,

$$\rho_{k\kappa}^{\mathrm{CQC}} = \frac{\mathrm{E}[q_k(t)q_\kappa(t)]}{\sqrt{\mathrm{E}[\{q_k(t)\}^2]\cdot\mathrm{E}[\{q_\kappa(t)\}^2]}}, \tag{77}$$

where $q_k(t) = h_k(\omega) * \ddot{u}_g(t)$ is the response of the kth mode.
- To determine FRS, the response of an oscillator (with frequency ω_0 and damping ratio ζ_0) mounted on the multiple DOF structure is required. The correlation coefficient between the responses of the oscillator contributed by two modal responses is determined for FRS-CQC, i.e.,

$$\rho_{k\kappa}^{\mathrm{FRS-CQC}} = \frac{\mathrm{E}[Q_k(t)Q_\kappa(t)]}{\sqrt{\mathrm{E}[\{Q_k(t)\}^2]\cdot\mathrm{E}[\{Q_\kappa(t)\}^2]}}, \tag{78}$$

where $Q_k(t) = \omega_0^2\omega_k^2 h_0(t) * h_k(t) * \ddot{u}_g(t)$ is the response of the oscillator contributed by the kth mode.
- When $\omega_0 \to \infty$, i.e., when the oscillator is very rigid, $Q_k(t) \to q_k(t)$. Therefore, $\rho_{k\kappa}^{\mathrm{FRS-CQC}}$ includes
  - the correlation between $Q_k(t)$ and $Q_\kappa(t)$,
  - the correlation between $Q_k(t)$ and $Q_\kappa(t)$,
  - the correlations between $Q_k(t)$ and $Q_\kappa(t)$ and between $Q_\kappa(t)$ and $Q_k(t)$.
- It is important to note that CQC was derived for responses of multiple DOF (Der Kiureghian, 1980, 1981), considering only the correlation between two modal responses $Q_k(t)$ and $Q_\kappa(t)$. If CQC (with $\rho_{k\kappa}^{\mathrm{CQC}}$) or SRSS is used in modal combination to generate FRS, large errors may occur, especially for structures with closely-spaced modes.

4. Numerical examples

The accuracy and efficiency of the direct spectra-to-spectra method developed in Section 3 for generating FRS is demonstrated through numerical examples by comparing results from the proposed direct method with those from time history analyses.

The primary source of variability in time history analysis stems from the inherent uncertainties and randomness of the time histories, reflected in the rugged spectral shapes of FRS. Although there are large variations in individual FRS as shown later in this section, the statistical results (such as mean FRS, median FRS, or FRS with 84.1% non-exceedance probability (NEP)) from a number of time histories converge to smooth spectra as the number of time histories increases.

In this study, the statistical results of FRS of a large number of time history analyses are considered as the "exact" FRS and used as benchmark for verifying the accuracy of the direct method.

4.1. Model information

A service building of a nuclear power plant is selected as the primary structure. A three-dimensional finite element model of the building, as shown in Fig. 7, is established using the commercial finite element analysis software STARDYNE.

The superstructure of the building consists of steel frames and concrete floor slabs, and the basement is constructed using concrete. The elevation of each floor and the dimensions of the building are shown in Fig. 7. Some information of the finite element model is listed in Table 1.

Table 1
Information of finite element model.

	Node	Lumped mass	Beam		Shell	
			Element	Section	Element	Section
Number	1351	120	1740	31	830	8

Table 2
Modal information at Node 1.

Mode	Frequency (Hz)	Participation factor	Modal shape	Contribution factor
2	2.676	−7.413	−0.05082	0.38
20	5.838	−2.945	−0.02603	0.08
21	5.918	2.943	0.06409	0.19
31	7.212	−8.883	−0.01942	0.17
103	22.95	−100.8	0.00088	−0.09
106	23.96	−337.3	0.00024	−0.08

Table 3
Modal information at Node 2.

Mode	Frequency (Hz)	Participation factor	Modal shape	Contribution factor
2	2.676	−7.413	−0.14630	1.08
20	5.838	−2.945	−0.01904	0.06
21	5.918	2.943	0.04151	0.12
31	7.212	−8.883	0.03847	−0.34
105	23.34	−96.07	−0.00045	0.04
106	23.96	−337.3	−0.00011	0.04
107	23.98	−50.65	−0.00092	0.05

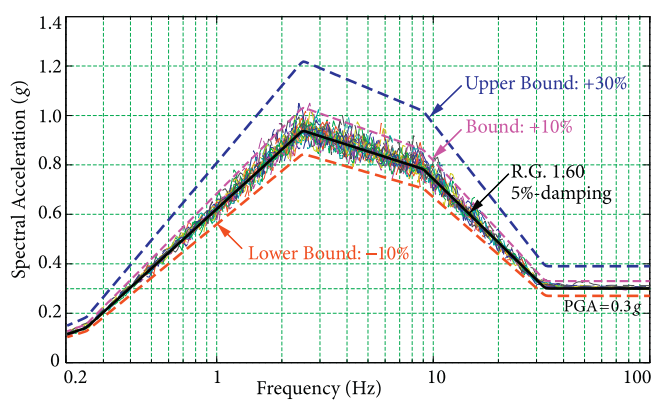


Fig. 8. Ground response spectrum.

A modal analysis is performed to obtain modal frequencies, modal participation factors, and modal shapes of the model. Modal information of 145 modes, in which the modal frequencies are less than 33 Hz, is extracted.

FRS at two nodes located on the second and third floors of the building are considered. Node 1 is on an edge of the second floor and Node 2 is on the third floor, as shown in Fig. 7.

Modal information of the significant modes at these two typical nodes is listed in Tables 2 and 3. The participation factors and modal shapes in these two tables are for direction 2 shown in Fig. 7. The contribution factor is the product of the participation factor and the modal shape, quantifying the contribution of the corresponding mode in the response of the node; all other modes that are not listed in Tables 2 and 3 have absolute values of the contribution factors less than 0.04. The summation of the 145 mode contribution factors at each node is close to 1. It is seen that there are closely spaced modes with considerable contributions to the responses at both Nodes 1 and 2. For example, modes 20 and 21 are closely spaced for Node 1; modes 20 and 21, modes 105–107 are closely-spaced for Node 2.

4.2. Input GRS

Two types of response spectra are selected as input GRS in the numerical examples.

GRS of USNRC R.G. 1.60. The 5% horizontal and vertical design spectra in USNRC R.G. 1.60 (1973) are taken as GRS in this example. The horizontal GRS are anchored at 0.3*g* PGA, and the vertical PGA is taken as 2/3 of the horizontal PGA. To perform time history analyses, 30 sets of tri-directional time histories spectrum-compatible with GRS (two horizontal GRS and one vertical GRS) are generated following the Approach 2 of USNRC SRP 3.7.1, as shown in Fig. 8.

Standard UHS for CENA. The 5% standard Uniform Hazard Spectrum (UHS) proposed by Atkinson and Elgohary (2007) for Central and Eastern North America (CENA) sites anchored at 0.3*g* is chosen as the horizontal GRS; the vertical input GRS is taken as 2/3 of the horizontal GRS. 30 sets of time histories compatible with the standard UHS are generated following the requirements of CSA N289.3 (2010), as shown

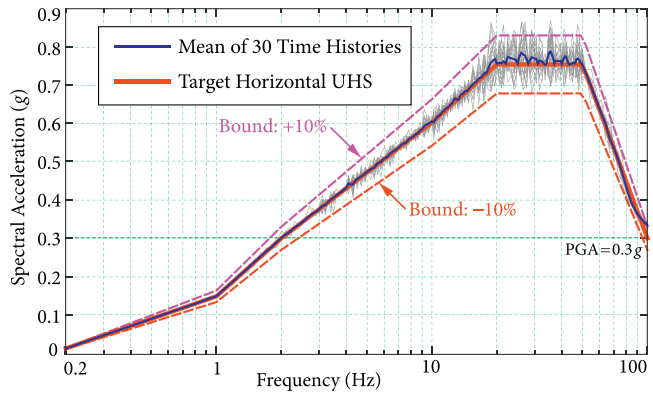


Fig. 9. Response spectra of compatible time histories.

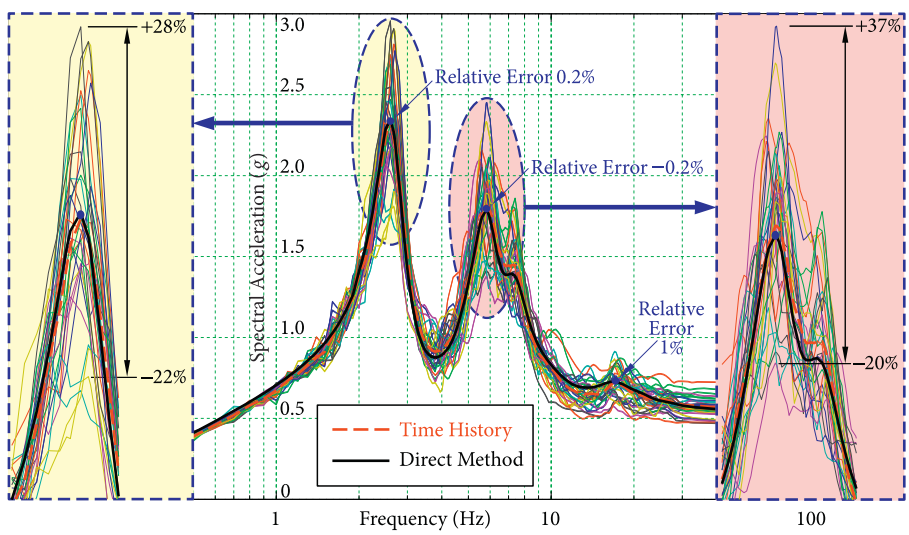


Fig. 10. FRS for Node 1 (USNRC R.G. 1.60 GRS).

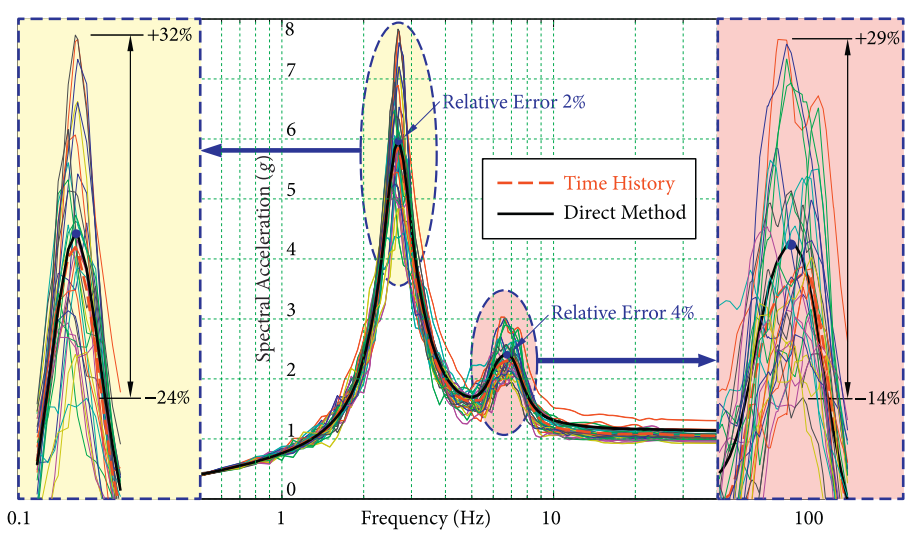


Fig. 11. FRS for Node 2 (USNRC R.G. 1.60 GRS).

in Fig. 9. Compared to design spectrum in USNRC R.G. 1.60, the most apparent characteristic of the standard UHS is the different spectrum shape; the standard UHS has relatively low spectral acceleration values in the intermediate frequency range from 2 to 10 Hz, but contains abundant higher frequency content in the frequency range from 20 to 50 Hz.

For both USNRC R.G. 1.60 GRS and CENA UHS, all spectrum-compatible time histories are generated using the Hilbert–Huang Transform method (Ni et al., 2011, 2013).

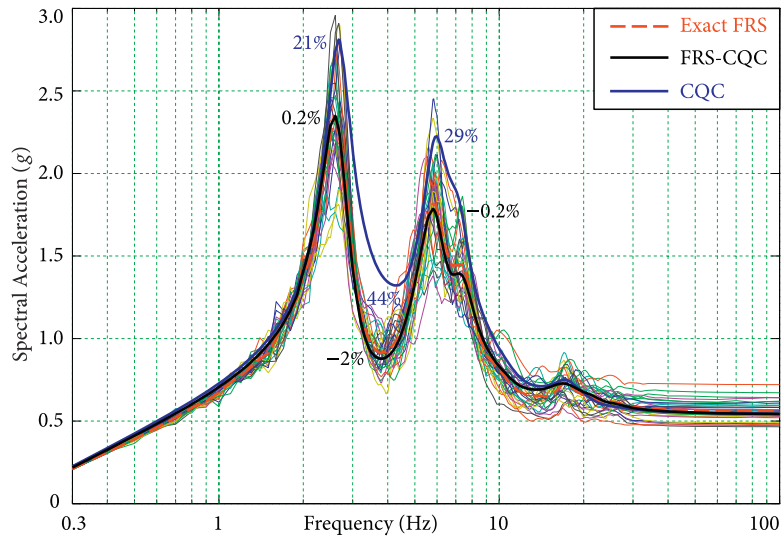


Fig. 12. Errors in modal combination rules for FRS at Node 1.

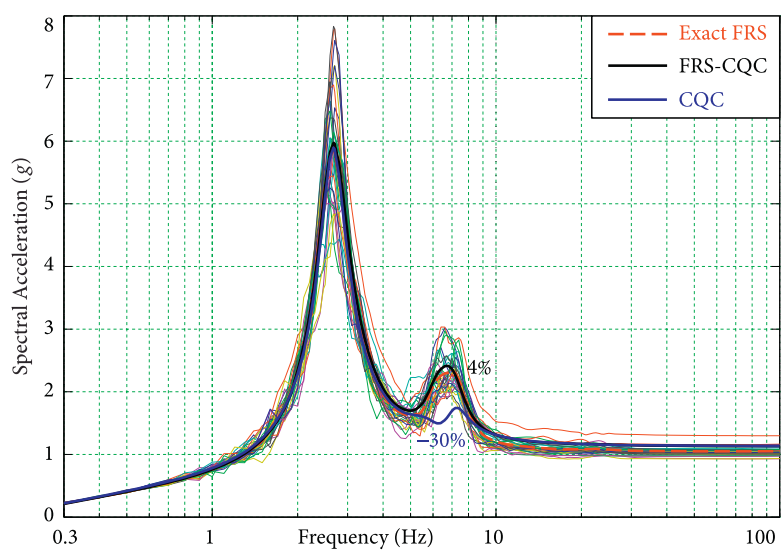


Fig. 13. Errors in modal combination rules for FRS at Node 2.

4.3. Comparison of FRS

FRS of the service building under the excitation of GRS of USNRC R.G. 1.60. FRS at Node 1 and Node 2 obtained from both time history analyses and the direct spectra-to-spectra method are plotted in Figs. 10 and 11, respectively. These FRS are calculated over 200 frequencies including natural frequencies of the dominant modes of the structure. The mean FRS obtained from time history analyses, which are considered as the "exact" FRS, are highlighted by red dashed lines; the FRS generated by the proposed direct method are represented as black solid lines.

It is seen that FRS generated by the direct method agree extremely well with the "exact" FRS over the entire frequency range. The relative errors are less than 5% at the peaks of FRS, whereas there are large variabilities in FRS from time history analyses, as shown in Figs. 10 and 11. This example demonstrates that time history analysis can lead to approximately 30% overestimation or 20% underestimation at the FRS peaks, even though the time histories are well compatible with the target GRS (within 10% of the target GRS). Furthermore, it is observed that FRS from a single time history analysis may be over-conservative at some peaks but significantly nonconservative at other peaks.

The primary source of variability in time history analysis stems from the inherent uncertainties and randomness of the spectrum-compatible time histories, which are reflected from their rugged spectral shapes. As seen in Figs. 8 and 9, there is an apparent difference between the response spectrum of a spectrum-compatible time history and the target GRS, which has a smooth spectral shape. From Eq. (38), it is clear that FRS are amplified GRS. Therefore, this difference is also amplified by an amplification factor, which can range from 5.5 to 7 in tuning cases. For an oscillator mounted on a SDOF structure, a 5% difference in GRS can result in approximately 30% difference at FRS peaks. For an oscillator mounted on a multiple DOF structure, modal responses are multiplied by the contribution factors and combined through Eqs. (60) and (76). As a result, variabilities in FRS are combinations of the differences in all modal response and are also significant.

To further verify the significance of modal combination on the determination of FRS, Figs. 12 and 13 show the FRS generated by the direct method using the conventional CQC combination rule. For FRS at Node 1, there are 21%, 29%, and 44% relative errors at the first,

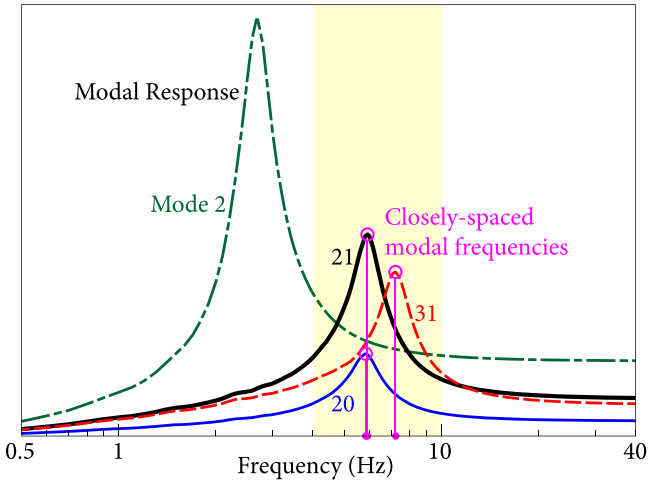


Fig. 14. Analysis of modal combination of FRS.

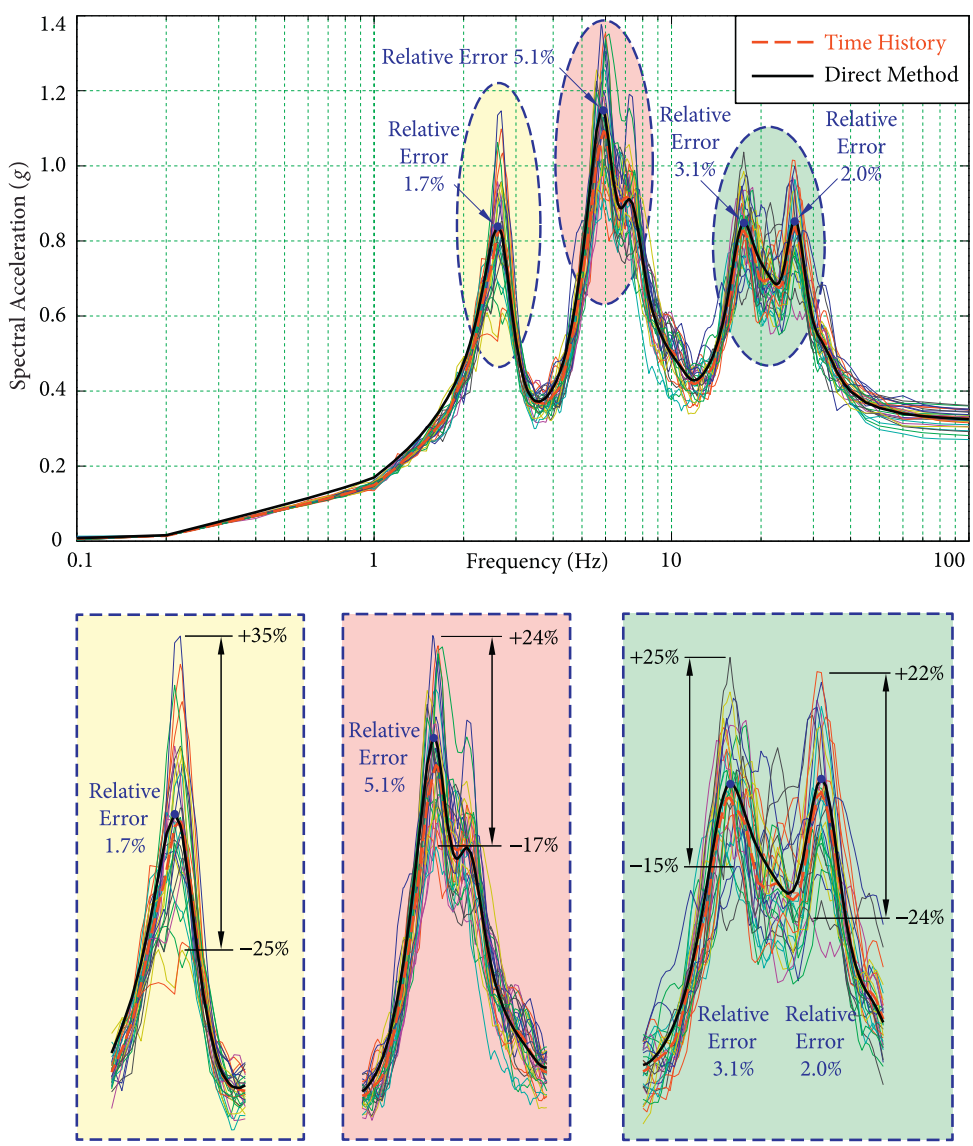


Fig. 15. FRS for Node 1 (UHS for CENA).

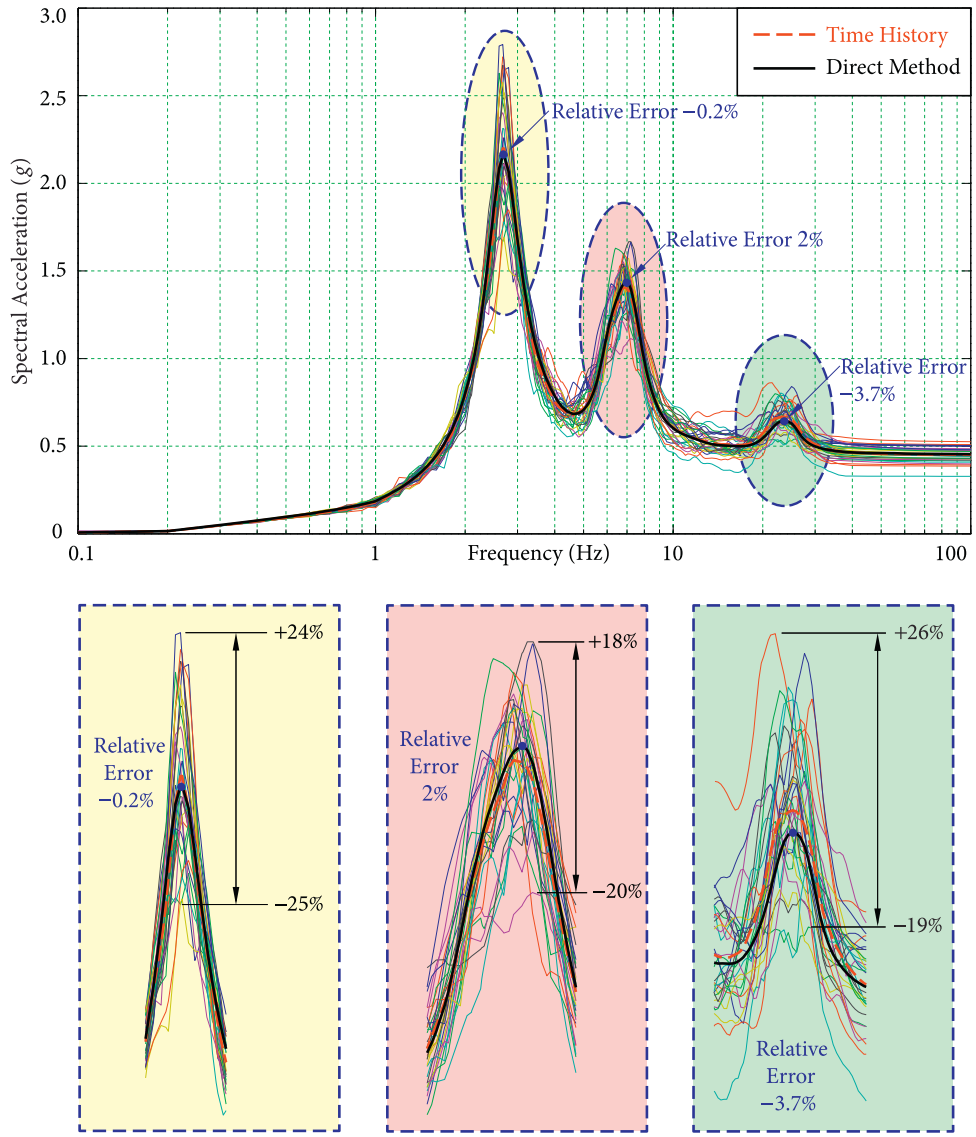


Fig. 16. FRS for Node 2 (UHS for CENA).

second peaks, and the valley between the two peaks, respectively. For Node 2, CQC gives results close to the "exact" FRS around the first peak. However, there is 30% under-estimation at the second peak.

Fig. 14 illustrates the seismic responses of the first four significant modes at Node 1. The effect of modal combination can be analyzed qualitatively as follows.

- For low frequencies $f < 2$ Hz, the response of the oscillator is contributed mainly by the amplified ground motion. Since the oscillator is a SDOF system, it does not involve modal combination.
- For frequencies from 2 to 20 Hz, which cover the dominant modal frequencies of the structure, each mode has considerable contribution so that the effect of modal combination becomes significant. Conventional CQC combination rule, which is developed to combine structural responses to account for the correlation between modal responses of the structure, cannot fully account for the correlation between the responses of the oscillator contributed by different modal responses and the correlation between response of the oscillator contributed by a modal response and the response of a structure mode.
- For high frequencies $f > 20$ Hz, because the oscillator is sufficiently rigid, its response is close to the structural response. The formula of FRS-CQC can be reduced to CQC in this case; hence, the resultant FRS given by these two combination rules are close.

Since structures in nuclear power plants have multiple dominant modes, and some of them are closely-spaced, modal combination rules significantly influence the resultant FRS. It is demonstrated that FRS-CQC combination rule is valid and accurate to combine modal responses.

FRS of the service building under the excitation of standard UHS for CENA. For Standard UHS for CENA, the proposed direct method is validated by comparing FRS obtained from the direct method with the "exact" FRS, as shown in Figs. 15 and 16. It is seen that FRS given by the proposed direct method agree extremely well with the "exact" FRS over the entire frequency range, and the relative errors at peaks are mostly less than 5%. Compared to Figs. 10 and 11, there are some differences in the spectral shapes of FRS, particularly over the higher

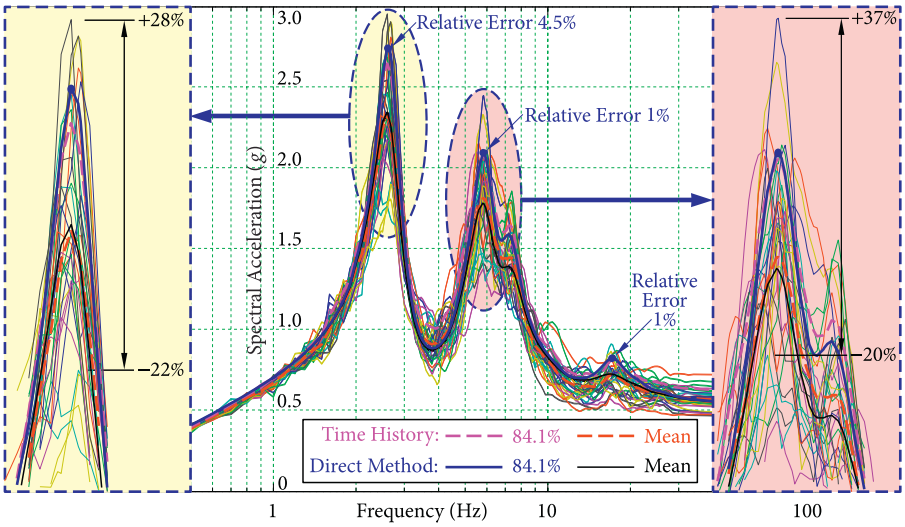


Fig. 17. Probabilistic description of FRS for Node 1 (USNRC R.G. 1.60 GRS).

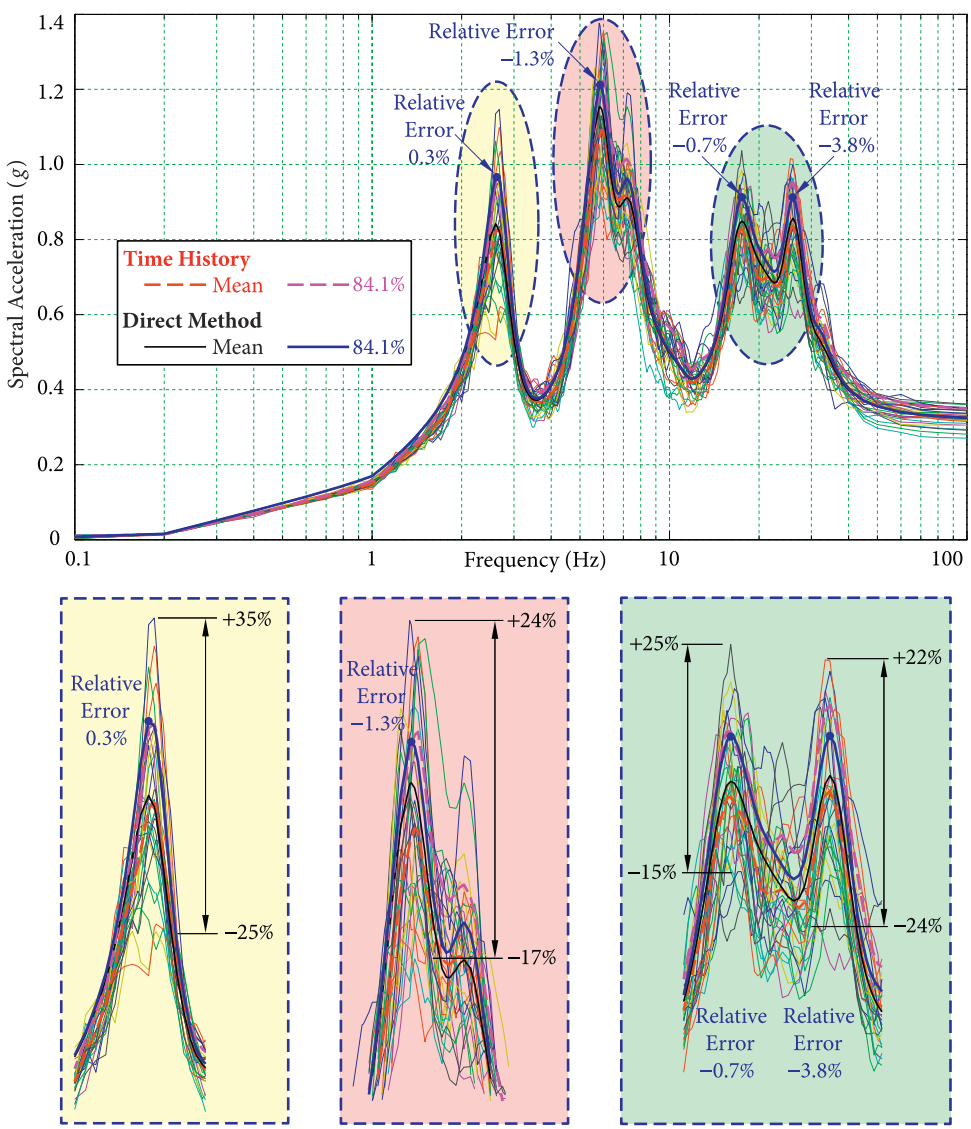


Fig. 18. Probabilistic description of FRS for Node 1 (UHS for CENA).

frequency range from 10 to 40 Hz. FRS peak up in this range since the spectral acceleration of UHS reaches the maximum value while the spectral acceleration of R.G. 1.60 GRS decreases.

Peak FRS generated from UHS are generally lower than those from R.G. 1.60 GRS. The reason is that the spectral accelerations of UHS are apparently lower than those of R.G. 1.60 over the frequency range from 2 to 8 Hz, where the dominant modes of the structure contribute most.

Probabilistic descriptions of FRS peaks. Another major advantage of the proposed direct method is that it can give probabilistic descriptions for FRS peak values. Using the probabilistic description of tRS developed in the companion paper, FRS with any desired level of NEP p can be determined from the given GRS and the corresponding tRS with NEP p. FRS with 84.1% NEP at Node 1 obtained by time history analyses and the proposed direct spectra-to-spectra method are compared in Fig. 17 for USNRC R.G. 1.60 GRS and in Fig. 18 for Standard UHS for CENA. The relative errors at the three FRS peaks are 4.5%, 1%, and 1%, respectively, for USNRC R.G. 1.60 GRS; the relative errors at the four FRS peaks are 0.3%, −1.3%, −0.7%, and −3.8%, respectively, for Standard UHS for CENA. This excellent degree of agreement further demonstrates the accuracy of the proposed method.

It is noted that the mean FRS and FRS with 84.1% NEP given by the proposed direct method are almost the same for non-tuning cases, as shown in Figs. 17 and 18. This can be explained by Eq. (43) and Fig. 5. The amplification factors given by Eq. (43) depend on the equivalent damping ratios $\zeta_{0,e}$ and ζ_e, which are determined by tRS and GRS in Eq. (54) and have a significant effect on the amplification factors in the perfect-tuning and near-tuning cases but have a negligible effect in non-tuning cases, as shown in Fig. 5. As a result, for non-tuning cases, the amplification factors are almost the same for all values of the equivalent damping ratios, leading to that FRS for all levels of NEP p are almost the same.

Because of the large amplification factors in the perfect-tuning and near-tuning cases, small deviations of the response spectrum of a time history from the target GRS are significantly amplified. Although the compatibility of the time histories is good by satisfying code requirements, as shown in Figs. 8 and 9, there are large variabilities in the FRS from time history analyses, particularly in the perfect-tuning and near-tuning cases, as shown in Figs. 10, 11, 15 and 16. Peak responses can be overestimated and underestimated by as much as 35% and 25%, respectively. Hence, results from time history analyses using a single set or a small number of sets of spectrum-compatible ground motions are not adequate to give accurate FRS. This observation further highlights the advantage of the proposed direct method, which uses the target GRS as input directly, without generating spectrum-compatible time histories that are the primary source of variabilities.

5. Conclusions

In this study, a direct spectra-to-spectra method is developed for accurate and efficient generation of FRS. Seismic response of a SDOF oscillator mounted on a SDOF primary structure is studied first. The response of the oscillator is derived analytically based on Duhamel's integral. For the tuning case when the oscillator is resonant with the SDOF primary structure, the concept of tRS is introduced, along with the statistical relationships between tRS and GRS developed in the companion paper, to give an accurate and complete probabilistic description of FRS, since large variabilities exist in this important case.

The formulation is then extended to a SDOF oscillator mounted on a multiple DOF primary structure. A new modal combination method for generating FRS, called FRS-CQC, is developed based on random vibration theory. FRS-CQC can fully account for the correlation of responses between equipment and its supporting structure with closely-spaced modes.

Numerical examples of a typical service building subjected to tri-directional seismic input are presented. FRS determined by the proposed direct method agree extremely well with the "exact" FRS, which are obtained through a large number of time history analyses, for both conventional Newmark-type GRS (USNRC R.G. 1.60 GRS) and UHS for CENA with significant high frequency spectral accelerations. It is demonstrated that FRS determined by time history analyses have large variabilities, particularly in tuning cases or at FRS peaks; hence, FRS determined by time history methods using a single set or a small number of sets of spectrum-compatible tri-directional time histories are not reliable. Furthermore, it is shown that modal combination methods significantly affect the results; there will be large errors if the conventional CQC or SRSS modal combination methods are applied to determine FRS. The proposed direct spectra-to-spectra method can avoid these deficiencies and give accurate FRS, especially probabilistic descriptions of FRS peaks.

The proposed direct spectra-to-spectra method has three significant and novel features:

1 Using the concept of tRS and the statistical relationships between tRS and GRS, FRS in the perfect-tuning and near-tuning cases can be determined accurately for both conventional Newmark-type GRS and UHS with significant high frequency spectral accelerations.
2 The correlations of responses between equipment and it supporting structure with closely-spaced modes can be fully accounted for through FRS-CQC combination rule. As a result, the proposed direct method can generate accurate FRS for complex three-dimensional finite element structural models with closely-spaced modes under tri-directional seismic input.
3 From the complete probabilistic descriptions of tRS for given GRS, the proposed direct method can give complete probabilistic descriptions of FRS peaks, i.e., FRS with any desired level of NEP p can be determined.

In summary, the proposed direct spectra-to-spectra method is accurate and efficient, with analytical formulation in terms of basic modal information of the primary structure, prescribed GRS, and the corresponding tRS. It can be conveniently implemented in practice to generate FRS for complex structures with closely-spaced modes.

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